Runaway edges: affine null spaces of smoothness penalties destabilise Kolmogorov–Arnold networks on correlated inputs

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Abstract

Kolmogorov–Arnold networks (KANs) are promoted as interpretable because each edge carries a learned univariate function. We show that a widely used family of smoothness penalties leaves a blind spot that undermines this promise: second-difference and curvature penalties do not charge for affine components of an edge, and SiLU base weights are often unpenalised. When the inputs to a trained nonlinear layer are correlated, affine components on different edges can cancel, so training inflates the parameters along directions that neither the penalty nor the data can see, and independent runs do not agree. In pre-registered experiments with a known teacher, correlated inputs gave 0 of 16 converged models, including under a recently proposed B-spline curvature penalty, while a small penalty on the affine null space gave 16 of 16 certified minima and essentially exact recovery of the teacher. Along certified regularisation paths spanning five decades, the loss behaves as if its infimum lies at infinity, on synthetic data and on real electrolyte-conductivity data; closing the null space bounds it. Non-convergence rises with input correlation, and the runaway and its fix carry over from Gaussian-RBF to B-spline edges. On the real data, polishing beyond early stopping degraded predictions for one held-out co-solvent, ethyl acetate (RMSE 0.142 $\rightarrow$ 0.81 without a null-space penalty and 1.03 with a weak one; 0.33 with a stronger one). Certified minima can still be degenerate or distinct, so convergence and identifiability are separate problems. We give practical recommendations for penalised KANs.

1 Introduction

KANs replace a network's fixed activations with learned univariate functions on every edge. Their appeal for science rests on reading those functions: an edge that looks like an exponential decay suggests a physical law. That reading presumes the edges are determined by the data. If different training runs produce different edges with equally good predictions, the edges describe the optimiser, not the system.

This paper identifies a concrete, common mechanism that breaks that presumption. It arises from the interaction of three ingredients that are each standard in practice:

1. **A smoothness penalty with an affine null space.** Second-difference (P-spline) penalties and curvature penalties ignore straight-line components of an edge; SiLU base terms are frequently unpenalised.
2. **Correlated inputs to a layer.** Physical descriptors often move together. Hidden units can also become correlated during training, exposing the next layer.
3. **Training toward convergence.** Early stopping hides the problem; longer optimisation exposes it.

Together they create directions along which edges grow into large, mutually cancelling affine shapes. The penalty does not see them, and on the training data their contributions cancel. The loss keeps decreasing

slightly along these directions, and over the range we could test it behaves as if it had no finite minimum. Seeds wander apart, no model certifies as converged, and on inputs outside the training data's support the cancellation can fail.

Contributions.

- We describe the mechanism: a penalty's affine blind spot plus correlated inputs to a trained nonlinear layer. A layer whose output is linear in its parameters cannot run away (Sections 3 and 6).
- In a pre-registered synthetic study, adding a SiLU-weight penalty in the style of a recent curvature penalty does not prevent the runaway on correlated inputs, nor does that penalty's exact B-spline form (Section 8); a small penalty on the affine null space prevents it and, in this well-specified setting, restores recovery of a known solution (Section 4).
- On real electrolyte-conductivity data, we trace a complete failure to converge to this mechanism and document its effect on a held-out co-solvent (Section 5).
- Along certified regularisation paths, the loss behaves as if its infimum lies at infinity over five decades of regularisation, on synthetic and real data; closing the null space bounds it (Section 6).
- Non-convergence rises with input correlation, and the runaway and its fix carry over from Gaussian-RBF to B-spline edges (Sections 7 and 8).
- Certified minima can be degenerate, distinct or not minima at all; we separate three kinds of flatness and give recommendations (Sections 9 and 10).

All confirmatory studies were pre-registered: predictions, thresholds and interpretation rules were committed before the code that tested them. Of 65 predictions in the real-data case study, 11 held, 22 failed and 32 were inconclusive; we report them as recorded. The revision studies of Sections 6–8 and the applicability analysis followed the same protocol; Appendix A lists their verdicts, and Appendix D every change made after an outcome was known.

Terms. We keep four phenomena apart: exact symmetries of the parameterisation, along which the loss is exactly constant and which carry no information; near-flat directions from poor conditioning; failures of the optimiser to finish, with solver errors reported separately from iteration caps; and distinct minima, which make the fitted function itself depend on the seed. "Certified" means a numerically certified local minimum under stated tolerances, not a global or unique one; Appendix G gives the tolerances and defines every metric used.

2 Background and related work

KAN edges. A KAN layer maps inputs to outputs by summing learned univariate functions, one per edge. In the original formulation (Liu et al., 2025) each edge is a SiLU base term plus a B-spline expansion. FastKAN (Li, 2024) approximates the B-splines with Gaussian radial basis functions on fixed centres, which is the edge type used in our experiments:

$$\varphi(x) = \alpha \,\mathrm{SiLU}(x) + \sum_{k=1}^{K} c_k \exp\left(-\left(\frac{x-\mu_k}{h}\right)^2\right) \tag{1}$$

Smoothness penalties and their null spaces. P-spline penalties charge squared second differences of the coefficients (Eilers & Marx, 1996). Constant and linear coefficient patterns cost nothing: the penalty's null space is affine. In generalised additive models this is well understood, and Marra & Wood (2011) proposed an additional penalty on the null space (the "double penalty") for variable selection; Wood (2017) treats null spaces and concurvity in additive models in depth. Our fix is the same construction; what is new is its consequence inside a KAN, where edges are summed within units and passed through further layers.

Curvature penalties for KANs. Bagrow (2026) proposed per-edge curvature penalties for compositional smoothness, combining the second-difference penalty with a term $K_{\text{silu}}\,\alpha^2$ on the SiLU weight, and noted that the penalty has affine functions as its null space by design; for Gaussian-RBF edges he instead uses the integrated bending energy, whose curvature Gram matrix charges near-affine coefficient patterns slightly. That work evaluated on independent, uniformly sampled inputs. Second-derivative penalties on spline coefficients also appear in other KAN variants, typically on the spline part only, leaving the SiLU base term free (Cang et al., 2024; see also the free-knots KAN of Zheng et al., 2025). We show that the affine null space becomes a runaway direction when inputs are correlated, and that penalising the SiLU weight alone does not close it.

Concurvity. In additive models, including neural additive models (Agarwal et al., 2021), correlated inputs let one input's shape function absorb another's, making shape functions non-identifiable; this is known as concurvity. Siems et al. (2023) proposed a regulariser penalising correlations between transformed features. The cross-edge cancellation we describe is concurvity inside a KAN unit. In an additive model with a penalised null space it shifts shape between functions; combined with an unpenalised affine null space in a KAN, it becomes unbounded growth.

Losses without a finite minimum. When a loss attains its infimum only at infinity, gradient descent drives the parameter norm to diverge, slowly, and optimisation can continue while held-out performance worsens (Soudry et al., 2018). Our runaway behaves like an instance of this, created by a penalty's null space. It differs from the benign flat directions of linear least squares, where gradient descent converges to the minimum-norm solution: here the direction is slightly tilted, so gradient descent keeps following it. On such ill-conditioned problems, early stopping acts much like ridge regularisation (Yao et al., 2007; Ali et al., 2019).

Parameter-space symmetries. Exact symmetries (unit permutations, sign flips) make parameters non-identifiable in tanh and RBF networks, and completeness results characterise them (Sussmann, 1992; Kůrková & Neruda, 1994). KANs add continuous symmetries such as affine reparametrisations of hidden units. The runaway is not an exact symmetry: it is an approximate one, a nearly flat but tilted direction, which is what prevents convergence.

3 Mechanism

Consider a unit that sums edges from two inputs, $s = \varphi_1(x_1) + \varphi_2(x_2)$, each edge as in Section 2, trained with a second-difference penalty on its RBF coefficients.

Step 1: a direction the penalty cannot see. Add a linear pattern $t\,\ell$ to the RBF coefficients of edge 1 and subtract it from edge 2, with ℓ_k proportional to the centre index. Linear coefficient patterns have zero second difference, so the penalty is unchanged for every t (Figure 1b). With Gaussian bumps on a regular grid, the pattern produces a nearly affine function: over the central 80% of the input range it deviates from a straight line by at most 2% of its range (Figure 1a), and by up to 13% near the grid edges. Edge 1 thus gains approximately $t\,a\,x_1$ and edge 2 loses $t\,a\,x_2$.

Step 2: a direction the data cannot see. The unit's pre-activation changes by approximately $t\,a\,(x_1 - x_2)$. If x_1 and x_2 are strongly correlated, $x_1 - x_2$ is small wherever the data lie, so predictions change little even for large t (Figure 1c). The direction is nearly free in both the penalty and the data.

Step 3: a receding minimum. "Nearly free" is not "free". Moving along the direction lets the model shed curvature into the free affine and SiLU components, so the loss keeps decreasing slightly. Both terms contribute (post-hoc decomposition): in the synthetic runaway models the smoothness penalty fell to about a quarter of its early value and supplied a median 61% of the decrease while training error fell only 5%; on real data, from early stopping to the capped state, the penalty fell 8–88× and supplied 17–36% of the decrease, the data term the rest. The growth went into whatever the penalty left free: the linear part of the RBF coefficients grew 34–40× in the synthetic runaway cells, and the SiLU weights 85× where they were unpenalised. In 12 of 20 synthetic runaway models, training error fell while test error rose. Over the range we could test, the loss behaves as if its infimum lies at infinity: along a certified L2 regularisation path, the minimisers grow without levelling off over five decades of the regularisation weight while the loss

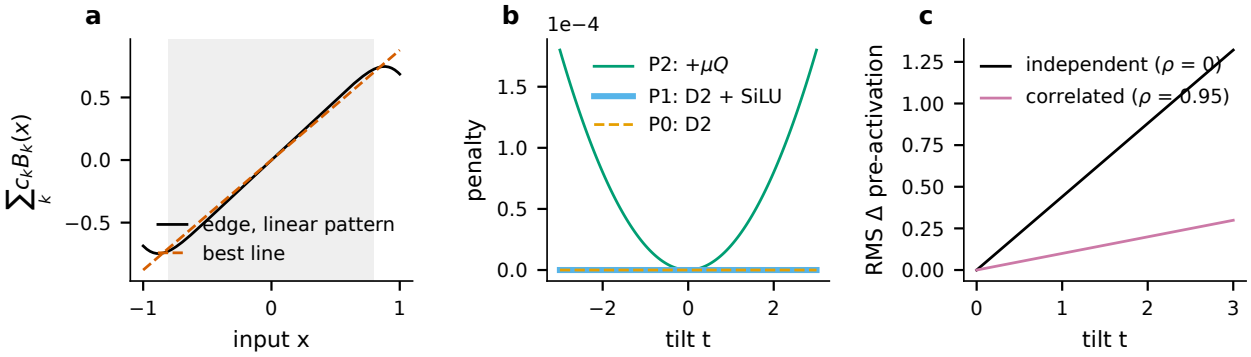


Figure 1: Mechanism (computed illustration) for the edge type of Section 2 (8 Gaussian RBFs on $[-1, 1]$). **(a)** Edge function produced by a unit linear coefficient pattern and its best straight line; the shaded band is the central 80% of the range, where the maximum deviation is 2.0% of the function's range (12.9% over the full range). **(b)** Penalty as a function of a tilt t applied as $+t$ to edge 1 and $-t$ to edge 2: the second-difference penalty (P0) and P1 are exactly zero for every t; only P2's term μQ grows ($\lambda = 10^{-3}$, $\mu = 10^{-5}$). **(c)** RMS change in the unit's pre-activation per unit tilt: 0.0996 on correlated inputs ($\rho = 0.95$) against 0.4403 on independent inputs (4.42×).

keeps falling, and they recede along a curved valley rather than a straight ray (Section 6). Growth alone is not the signature, because a distant finite minimum also produces large growth (Step 4); what distinguishes the runaway is that the optimisation never reaches a certified minimum. Different seeds drift along different such directions.

Step 4: the runaway needs a trained nonlinear layer. If the parameters being trained enter the output linearly, as in the output layer when the hidden activations are held fixed, or in a one-layer KAN, the penalised loss is a convex quadratic. It may have flat directions, but no tilted ones, so it cannot run away: with the first layer frozen, every model reached a certified finite minimum, and so did a one-layer model on correlated inputs, although the minimum could lie far away (growth up to ×583 on real data). The runaway therefore needs trained parameters that act through a nonlinearity. Correlation can reach such a layer from correlated inputs, or from hidden units that become correlated during training, so the output layer can host the runaway even when the raw inputs are independent, as long as the first layer also adapts. Correlated hidden units are necessary but not sufficient: they also occur in models that converge once the null space is penalised. Whether a given nonlinear problem degenerates depends on the data: with the second layer frozen, the synthetic first-layer problems settled cleanly, while the real first-layer problem, with few-valued, strongly correlated descriptors, reached only highly degenerate minima, one of which was not a minimum at all.

Consequences.

- **Non-identifiability:** edges differ across seeds by large affine components that leave predictions almost unchanged; even certified minima can be degenerate or distinct from one another.
- **Non-convergence:** second-order optimisers stall and certification fails. Since growth is also large on the way to a distant finite minimum, and norms depend on how the model is parameterised, certification and function-space checks, not growth, are the criteria.
- **Extrapolation risk:** on inputs outside the training data's support, the inflated components need not cancel. In our real-data case, polishing damaged predictions for a held-out co-solvent lying outside the training descriptor hull.

The fix. Penalise the null space directly. For each edge, let P_{null} project its RBF coefficients onto constant-plus-linear patterns in the centre index, and add

Table 1: Synthetic 2×3 study, per cell.

Cell	Norm growth	Certified	Unit match to reference
Independent, P0	12.5×	6/8	1.000
Independent, P1	7.9×	6/8	1.000
Independent, P2	1.09×	8/8	1.000
Correlated, P0	65.7×	0/8	0.774
Correlated, P1	32.1×	0/8	0.883
Correlated, P2	1.00×	8/8	1.000

$$Q = \sum_e \|P_{\text{null}}\, c_e\|^2 \tag{2}$$

with a small weight μ, in addition to a penalty on the SiLU weight. This makes every affine component cost something, so the regularised minimisers stay bounded. Penalising only part of the free directions is not enough: the growth moves into whatever stays unpenalised, such as biases or embeddings, so these need a small penalty too.

4 Controlled study

Design. A teacher KAN [4, 3, 1] with smooth, hand-specified edges (fitted by smoothness-penalised least squares) generates targets with Gaussian noise of 5% of the output spread; n = 2000 training rows. Inputs are $\mathcal{N}(0,1)/3$, clipped to the grid, either independent or correlated in pairs ($\rho = 0.95$ for x_1/x_2 and x_3/x_4). Students share the teacher's architecture and are trained under three penalty levels:

- **P0:** second-difference penalty on RBF coefficients ($\lambda = 10^{-3}$).
- **P1:** P0 plus $K_{\text{silu}}\,\alpha^2$ on each SiLU weight ($K_{\text{silu}} = 0.443$; Bagrow's form for B-splines, but only a surrogate for his Gaussian-RBF penalty, which uses a curvature Gram matrix; Section 8 tests his B-spline form exactly).
- **P2:** P1 plus μQ, the affine null-space penalty.

Each of the 6 cells has 8 seeds. Each student is trained with AdamW and early stopping, then polished with an exact-Hessian trust-region method and certified by a second-order criterion (gradient, curvature and Newton decrement). μ was calibrated before the experiment by a pre-registered rule: 10^{-6} failed the pilot (5.0× growth) and 10^{-5} passed (1.32×) and was frozen.

The teacher itself runs away. Polished under P0 and P1 on correlated inputs, the teacher's parameters grew 85× and 1069× without certifying, while its predictions stayed within half a noise standard deviation of the original. The runaway changes the parameters, not the function.

Medians over 8 seeds; growth is the squared parameter norm, polished over early-stopped.

Results.

- **Penalising the SiLU weight does not prevent the runaway.** P1 halved the growth on correlated inputs but left 0 of 8 models certified (Appendix E, Figure 5).
- **Closing the null space prevents it everywhere.** P2 gave 16 of 16 certified models with no growth, in both input conditions.
- **Without the runaway, the internals are recoverable.** In this well-specified setting, where teacher and student share architecture, basis and grid, every cell without a runaway matched the reference's units and edges exactly (1.000), with seed-to-seed differences around 10^{-4}.

- **Independent inputs are not fully safe.** Under P0 and P1, most independent seeds grew substantially (8–12×) but settled at certified minima; 4 of 16 (seeds 0 and 4 in each cell) never certified.

Where the runaway lives. A post-hoc anatomy shows that in the correlated cells almost all growth sits in the output layer (layer-1 share 0.110 under P0, 0.024 under P1; Appendix E, Figure 6a), in the affine part of the output edges' coefficients and, under P0, their SiLU weights, with no hidden unit saturating: Step 4 of Section 3. Among independent-input models, those that ran away were exactly those whose hidden units had become correlated (largest pairwise correlation 0.70, against 0.05 for certified models; Appendix E, Figure 6b), while certified P2 models kept correlations up to 0.88 without running away (post hoc).

Verdicts. Of six pre-registered predictions, one held (the certification pattern, X2) and five failed. Independent cells exceeded the growth threshold, mostly through finite growth. Cancellation and extrapolation effects were weaker than predicted. Recoverability failed only because the runaway cells were more recoverable than predicted, not less. In this synthetic setting, polishing did not degrade extrapolation: most of the off-pattern error in correlated cells was present at early stopping.

5 Real-data case study: electrolyte conductivity

Task and model. The CALiSol-23 challenge (de Blasio et al., 2024) asks for the log conductivity of lithium-ion electrolytes; every test sample contains diethyl carbonate (DEC), which never appears in training. KAN-δ anchors each prediction on the measured conductivity of a pure propylene-carbonate (PC) electrolyte with the same salt, from the same article (Appendix H), and learns only the correction, as in Δ-machine learning (Ramakrishnan et al., 2015), $\log k = \log k_{\mathrm{PC}} + x_{\mathrm{co}}\, h(z)$, where h is a FastKAN-style network of shape [10, 6, 1]. Its inputs are physical descriptors of the solvents (dielectric constant, log viscosity, molar mass), temperature, molality and a salt embedding. A second-difference penalty acts on the RBF coefficients, and the SiLU weights are unpenalised. Early-stopped after 107 AdamW epochs, it reached RMSE 0.142 on a held-out co-solvent (fold A: ethyl acetate held out). Its AdamW weight decay played no part: retraining without it left the early-stopped norms essentially unchanged and fold A's RMSE at 0.1420 (a two-sided pre-registered prediction whose verdict was the branch "incidental"). We use these data as an optimisation stress test with physical guidance, not as a transport model: the anchor and descriptors add structure, but the learned edges are not physical laws, and the salt embedding only interpolates among the salts seen in training.

The inputs are correlated and few-valued. The three co-solvent descriptors take only 30 distinct combinations in training, and dielectric constant and log viscosity correlate at r = 0.966.

No tested run met the certification criterion.

- Of 190 pairs of polished independent seeds, none computed the same function.
- After 20,000 float64 L-BFGS iterations (the "capped" state used below), 0 of 26 models converged; exact-Hessian trust-region methods also failed on every seed.
- Students trained on a frozen teacher matched its predictions (correlation 0.9988) but not its hidden units (0.66).

The anatomy matches Section 3.

- From early stopping to the capped state, the squared parameter norm grew 150–1550× while the smoothness penalty fell 8–88×.
- At least 99.9% of the coefficient change during exact-Newton polishing lay in the penalty's affine null space, in 9 of 9 models.
- Cancellation within an edge (SiLU against RBF) stayed small, 1.13–1.33. Cancellation across the edges entering a unit rose from 1.7–2.9 to 4–10, consistent with the correlated descriptors.

- The runaway spread across both layers: 30–40% of the parameter change went into layer-1 affine RBF coefficients, 37–38% into SiLU weights and 19–31% into layer-2 affine coefficients (post-hoc), and 12–39% of hidden units saturated. In the synthetic runaway the output layer dominated and no unit saturated, so saturation is not needed for the mechanism.

Extrapolation suffered. Polishing beyond early stopping raised fold A's RMSE from 0.142 to 0.81 with no null-space penalty and to 1.03 under a weak one ($\mu = 10^{-6}$); a stronger one ($\mu = 10^{-4}$) held it at 0.33 (Section 9). Fold A was used repeatedly during development, so it is not an untouched test set: the 107-epoch stopping point was chosen using its own held-out rows (and fold B's), which favours the early-stopped model here, and its 635 rows are one salt from one article (Appendix H). Fold A's co-solvent lies outside the training data's descriptor hull (Appendix E, Figure 7c), although on the correlation pattern of dielectric constant and viscosity. Within the fold, the damage scales with the co-solvent fraction: the learned correction enters as $x_{\mathrm{co}}\, h$, so any change polishing makes to h is magnified where x_{co} is large (descriptive; Appendix E, Figure 7a, b). Decomposing predictions into edge contributions shows the cancellation directly: on fold A a larger share of the opposing output-edge contributions survives than on training rows, already at early stopping (net-to-total ratio 0.59 against 0.31) and after polishing (0.12 against 0.06; post hoc, Appendix B). Across the co-solvent folds with at least 20 test rows, no distance measure distinguishably predicted where errors were large, and a planned predictive follow-up was dropped because its pre-stated precondition was not met. The degradation was much stronger than in the synthetic study, where correlation itself caused most of the off-pattern error (Section 11).

6 Is the minimum at infinity?

Failing to converge does not by itself show that the loss has no finite minimum. The minimum could be finite but very far away, or the problem could simply be too ill-conditioned for the optimiser to finish. Two pre-registered experiments separate these possibilities: freezing one layer, and following a certified regularisation path.

Freezing a layer. With the first layer frozen at its early-stopped weights, only the output layer is trained. Its parameters enter the output linearly, so the penalised loss is a convex quadratic. Every model then reached a certified minimum: 6 of 6 real KAN-δ seeds and 8 of 8 in each correlated synthetic cell, with no negative curvature at any logged iterate; a one-layer KAN trained on correlated inputs also certified, 8 of 8. The pre-registered prediction that the norm would grow at most threefold failed: the real models grew a median $\times 583$ to reach their minima. That growth is not a runaway, because it ends at a certified point. Freezing the second layer instead, in an exploratory arm, left the synthetic models certifying but the real first layer reaching only highly degenerate endpoints (Appendix F.1). Section 3 draws the conclusion: the runaway needs trained parameters acting through a nonlinearity, and whether such a problem degenerates depends on the data.

A certified regularisation path. For a loss L that is bounded below and a penalty R that grows without bound with the parameters, the global minimisers of $L + \gamma R$ stay bounded as $\gamma \to 0$ if and only if L has a finite minimiser (Appendix F.6). We use $\mathrm{R} = \|\theta\|^2$ over all parameters, follow γ from 10^{-2} down to 10^{-7} in half-decade steps, warm-start each value from the previous solution, and certify every point; degenerate points are accepted only after probes for remaining descent. On synthetic data, with the blind spot open, the minimisers' squared norm grew as a power of $1/\gamma$ (exponent 0.63–0.76 under P0, 0.36 under P1) while the unregularised loss kept falling; with the null space closed (P2), the path levelled off (exponent 0.014, ratio 1.004 between the two smallest γ). On the six real models, the same divergence appeared on six distinct branches: exponents 0.74–0.88, squared norms reaching 596–3052, and the unregularised loss falling from $2.05{\cdot}10^{-2}$ to 8.7–$9.3{\cdot}10^{-3}$. All pre-registered predictions held, the real-data discrimination check being taken from the synthetic controls as pre-registered. (Figure 2)

Two details matter for how far these replicate. On synthetic data the warm-started paths start from nearly one function; under P1 and P2 they stay together along the whole path, while under P0 they separate into a few branches. The real paths, by contrast, are genuinely distinct and compute different functions: at $\gamma \leq 10^{-4}$ their predictions differ by a median 13.7–16.3% of the output's standard deviation.

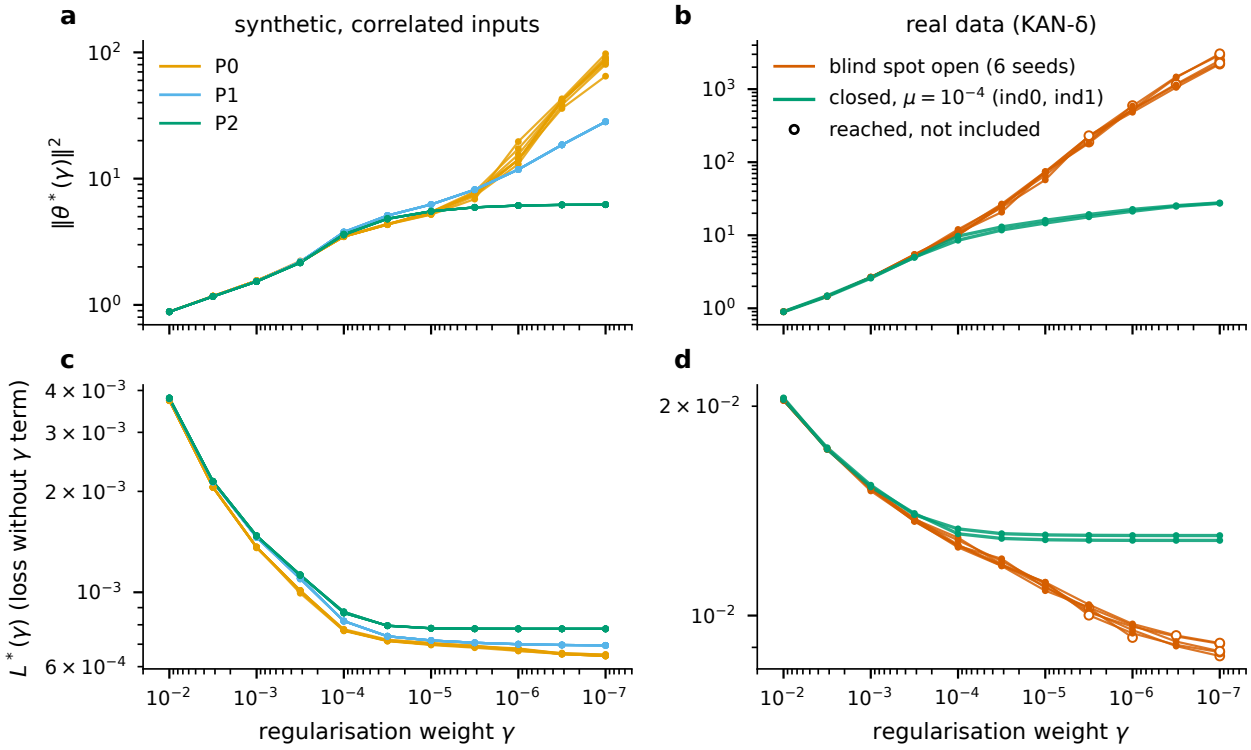


Figure 2: Certified L_2 regularisation paths, $\gamma = 10^{-2}$ to 10^{-7} in half-decade steps (warm-started), one line per seed. **(a, c)** Synthetic correlated inputs, 8 seeds per penalty level: $\|\theta^*\|^2$ grows as a power of $1/\gamma$ with exponent 0.63–0.76 (P0) and 0.36 (P1), and 0.014 with the null space closed (P2). **(b, d)** Real data: the 6 KAN-δ seeds with the blind spot open (exponents 0.74–0.88) and seeds ind0 and ind1 with the null-space penalty at $\mu = 10^{-4}$ (post-hoc control, outcomes fixed in advance; ratio between the two smallest γ 1.09 and 1.10). L^* is the loss without the γ term (for the closed paths, including μQ). Filled markers: points included in the pre-registered fits; open markers: reached but not included; absent: not reached.

Robustness (post hoc). Cold starts found no small-norm basin, and at the smallest γ often a lower objective at a larger norm, so the warm path understates the divergence; the minimisers recede along a curved valley, in the penalty's blind directions (Appendix F.7).

Closing the null space on real data. In a control whose two possible outcomes were fixed in advance, the same path on the real loss with the null-space penalty at $\mu = 10^{-4}$ levelled off: squared norms of 27.7 and 27.4 against 596 and 2200 for the same seeds with the blind spot open (ratios 1.09 and 1.10). The plateau is not a full stop: the norm still creeps up (whole-path exponents 0.105 and 0.124), in the biases, the curved RBF coefficients and the salt embedding, which this penalty leaves weakly penalised or free. The price is a higher training loss: +6% and +23% for the two seeds ($1.0926{\cdot}10^{-2}$ against $1.0300{\cdot}10^{-2}$, and $1.1224{\cdot}10^{-2}$ against $9.113{\cdot}10^{-3}$). When the penalty is instead lowered towards zero, the growth floods whichever parameters it leaves free: with the null-space penalty of Section 3 the salt embedding grows 44–218-fold, while a variant that also penalises biases and embeddings keeps the embedding bounded.

Reading. Over the tested range, $\gamma = 10^{-2}$ to 10^{-7}, the loss with the blind spot open behaves as if its infimum lies at infinity, on synthetic and real data alike, and closing the null space bounds it. Two limits apply: the criterion concerns global minimisers while the path follows one local branch, and a finite minimum lying beyond the tested range cannot be excluded.

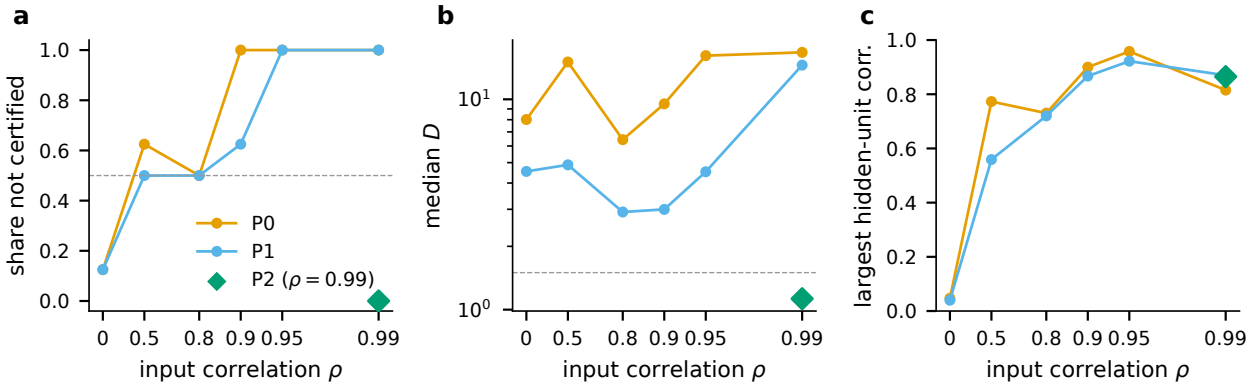


Figure 3: Correlation dose-response (synthetic [4, 3, 1], Gaussian inputs; 8 seeds per cell; the thresholds are synthetic, not real-data thresholds). Horizontal axis: ρ on a $\log 1/(1-\rho)$ scale. **(a)** Share of models not certified (dashed: one half). **(b)** Median short-path $D = \|\theta^*(10^{-7})\|^2/\|\theta^*(10^{-5})\|^2$ (log scale; dashed: 1.5). **(c)** Largest hidden-unit correlation, cell median. Diamonds: P2 at $\rho = 0.99$, 8 of 8 certified, median $D = 1.13$.

7 How much correlation does the blind spot need?

Section 4 compared two doses, independent inputs and $\rho = 0.95$. A pre-registered dose-response study fills in between: $\rho \in \{0, 0.5, 0.8, 0.9, 0.95, 0.99\}$ for both input pairs, P0 and P1 at every level, 8 seeds each, with paired random draws so that only the correlation changes, plus the null-space control P2 at $\rho = 0.99$. A replication gate ran first: the new $\rho = 0.95$ cells reproduced the committed synthetic study bit for bit.

Correlation drives non-convergence. The share of models that fail to certify rises with ρ (prediction held; P0 with one allowed inversion). It first exceeds one half at $\rho = 0.5$ under P0 and at $\rho = 0.9$ under P1, and from $\rho = 0.95$ no model certifies under either penalty. These thresholds belong to this synthetic setting ([4, 3, 1], Gaussian inputs); they are not thresholds for real data. With the null space closed, all 8 models certify even at $\rho = 0.99$, and a short regularisation path levels off (median D = 1.13, where D is the ratio of the squared norms at $\gamma = 10^{-7}$ and 10^{-5} along a short regularisation path). Along with the dose, the hidden units become correlated (largest pairwise correlation from about 0.05 at $\rho = 0$ to 0.82–0.96 at $\rho \geq 0.9$), and the polished training loss falls (Spearman -0.94): more correlation lets the runaway buy more loss. (Figure 3)

What a short path cannot tell. We also predicted that a short regularisation path ($\gamma = 10^{-3}$, 10^{-5}, 10^{-7}) would diverge more strongly as ρ grows. It did not: the path statistic D was already high at every dose, including $\rho = 0$. A post-hoc check with a rule fixed in advance showed why: where the polish certified, mostly at low ρ, 30 of 32 paths were heading towards that certified finite minimum, and a two-decade path cannot separate a distant finite minimum from a receding one (derivation in Appendix F.4). Receding behaviour (Section 6) is therefore shown only at $\rho = 0.95$ and on the real data, over the tested range; at low ρ the minima are finite but distant; at intermediate ρ these tests do not settle the question. A secondary prediction, that the affine share of the parameter change would rise steadily with ρ, also failed.

8 Is it the basis?

The experiments so far use Gaussian-RBF edges. To test whether the runaway belongs to the basis or to the penalty's blind spot, we replicated the synthetic 2×3 study with cubic B-spline edges as in pykan and efficient-kan: uniform knots with G = 5 intervals on $[-1, 1]$, extended by three knots at each end, giving 8 basis functions per edge (matching the 8 Gaussians), plus the SiLU term. On these knots, constant and linear coefficient patterns reproduce constant and linear functions exactly, so the second-difference penalty's blind spot is exactly the affine functions. The teacher was refitted into the B-spline basis; inputs, seeds and noise were those of the RBF study.

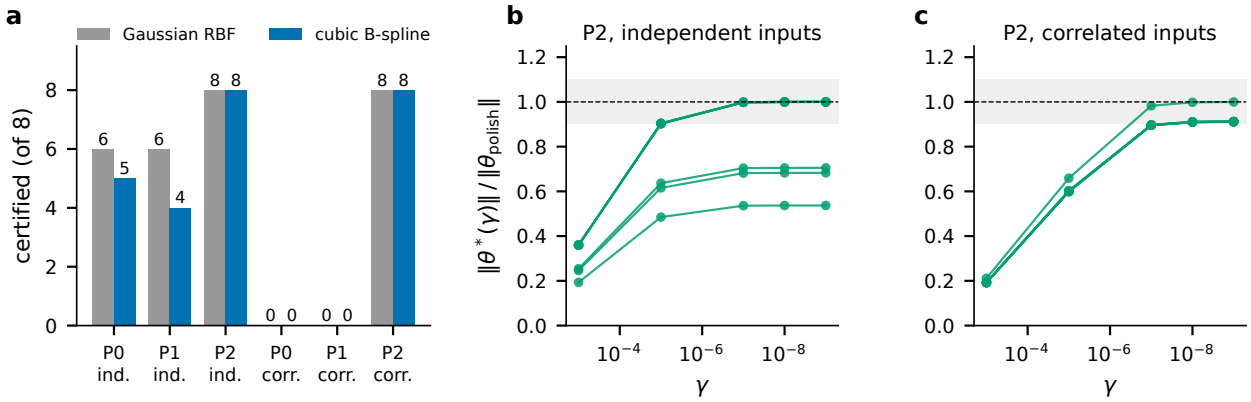


Figure 4: B-spline replication. **(a)** Certified models (of 8) per cell: the Gaussian-RBF study (frozen $\mu = 10^{-5}$) and the cubic B-spline replication (frozen $\mu = 10^{-6}$). **(b, c)** P2 regularisation paths at $\mu = 10^{-6}$, continued to $\gamma = 10^{-9}$ (post hoc, predictions fixed in advance): $\|\theta^*(\gamma)\|$ relative to each seed's certified polish minimum; shaded: the pre-fixed band [0.9, 1.1]. Median ratio at $\gamma = 10^{-9}$: 1.000 (independent) and 0.911 (correlated).

The runaway carries over. With correlated inputs, none of the 16 P0 and P1 models certified, where P1 here is Bagrow's B-spline penalty up to a scale factor our edges do not have (prediction held; Figure 4a). The runaway is not specific to the Gaussian basis: it follows the penalty's affine null space.

The fix carries over, with a lesson about calibration. The null-space weight was set by the same committed rule as in the RBF study (the smallest μ whose pilot models grew at most threefold), which here chose $\mu = 10^{-6}$, the bottom of the grid, against 10^{-5} for RBF edges. At that weight, the closed models certified 8 of 8 in both conditions, but the pre-registered bound on the short path failed with correlated inputs (median D = 2.23 against 1.5). Post-hoc checks, with predictions fixed in advance, traced this to the measurement window: the null-space penalty's curvature, about 1.8μ, lies between the two γ values that D compares. Extended to $\gamma = 10^{-9}$, the paths were bounded (Figure 4b, c), ending near the certified minima's norm though not always at the same point; and at the RBF study's $\mu = 10^{-5}$ the fix met the original standard (8 of 8 certified, D = 1.17 and 1.31). The growth-based rule chose a weight too small for a criterion carried over from another setting: a lesson about calibration, not about the fix.

Independent inputs. Fewer independent models certified than with RBF edges (5 and 4 of 8 under P0 and P1, against 6; one P1 failure was a solver error rather than an iteration cap), so the pre-registered prediction failed, and its fixed reading applies: the B-spline basis is less benign on independent inputs. Post hoc, two of the seven failures look like runaways, one like slow convergence, and four are mixed; with 8 models per cell, the margin is one or two models.

A symmetry the RBFs lack. B-spline edges carry an exact 15-dimensional symmetry, constants moved between a unit's incoming edges and its bias, which holds only approximately for RBF edges; we excluded these analytically constructed directions from certification and from the degeneracy count (Appendix F.5). This was one of five changes to the study's calibration step made after outcomes were known; none touched a prediction, a threshold or an interpretation rule, and all are listed in Appendix D.

9 Closing the blind spot, and what remains

The null-space weight on real data. Adding μQ (with the SiLU weights included in Q) and polishing from the early-stopped weights (Table 2).

Under a second-order certificate, all six seeds at $\mu = 10^{-4}$ were certified (Appendix E, Figure 8). (The originally pre-registered criterion, which required 10 stationary iterations, proved unreachable because the optimiser stops a few iterations after reaching a minimum; the certificate was defined afterwards and is

Table 2: Dose-response in μ on real data (seeds).

μ	Seeds at a numerical minimum	Negative curvature (seeds)	Norm growth (seeds)	Training fit vs runaway
10^{-6}	0 of 6	up to 9 eigenvalues	21–31×	−3% to +13%
10^{-5}	4 of 6	up to 4 eigenvalues	9–43×	12–22% worse
10^{-4}	6 of 6	none	1.6–16×	15–39% worse

reported alongside the original verdict.) Saturation fell to at most 0.5% and cross-edge cancellation to 2.2–2.9.

Against a regularised teacher, students recover its units and edges well, and the null-space penalty partly acts as gauge fixing (post hoc; Appendix F.3).

Convergence is not uniqueness. Independent seeds at certified minima still disagreed: their median prediction difference fell by only 26% (Appendix E, Figure 9b). With the runaway removed, the loss has several distinct isolated minima.

Three kinds of flatness. Exact symmetries are flat by construction and excluded analytically (Section 8). Near-flat directions from poor conditioning are not harmless: certified points with many of them can be degenerate, and one real-data point that passed the certificate kept descending on a longer restart while its squared norm grew 6.7-fold. We therefore report the number of near-flat directions with every certificate and probe degenerate points before counting them as minima. Certified is not unique, and not necessarily good: some certified minima lie at higher loss than points the regularisation path finds at smaller norm.

Extrapolation. On the held-out co-solvent, RMSE was 0.142 for the early-stopped model, 1.026 after polishing at $\mu = 10^{-6}$ and 0.33 at $\mu = 10^{-4}$. Bounding the runaway recovers most of the damage, but early stopping remains the most accurate choice here, though its stopping point was selected on this fold: on an ill-conditioned problem, stopping early keeps the poorly determined directions near their starting values, much as ridge regularisation would (Yao et al., 2007; Ali et al., 2019).

Structured alternatives. Two architectures designed to remove runaway routes by construction, a shared-curve KAN and a two-anchor excess model, were less accurate than the early-stopped baseline under leave-one-co-solvent-out validation; the shared-curve model converged every time but its seeds still disagreed (Appendix F.2).

10 Recommendations for practitioners

1. **Penalise the blind spot, and everything else that is free.** If a KAN uses a second-difference or curvature penalty, add a small penalty on the constant-plus-linear component of each edge's coefficients and on each SiLU weight, in every layer, and give biases and embeddings a small penalty too: penalising only part of the free directions redirects the growth into the rest. This restores bounded minima, not the accuracy of early stopping.

2. **Calibrate that penalty by certification and a plateau, not by growth.** Choose its weight so that models certify and the regularisation path levels off; a rule based on norm growth chose a weight too small in our B-spline replication.

3. **Check input correlations, including between hidden units.** Strongly correlated or few-valued inputs make a trained nonlinear layer vulnerable, and hidden units that become correlated expose the next layer.

4. **Judge convergence by certification, not by norms or the loss.** Large growth can end at a finite minimum; a runaway is growth that never reaches a certified one. Report the number of near-flat directions with every certificate, and probe degenerate points before calling them minima.

5. **Do not read edges from unconverged models.** Compare edges across seeds only after removing exact symmetries, and only for models at certified minima.

6. **Treat seed disagreement as information.** Even at certified minima, distinct solutions remain; disagreement then signals that the data do not determine the function, not merely that training was unfinished.

7. **Validate on genuinely new conditions, and keep early stopping as a baseline.** A held-out fold from the training distribution can understate extrapolation error severalfold; in our real-data case, early stopping gave the best held-out accuracy of every variant tested, though its stopping point was selected on that fold. When extrapolative accuracy matters most, keep it; when the aim is to read edges, close the null space, accept its training-loss cost, and compare certified seeds.

11 Limitations and research process

Scope.

- One real dataset and one synthetic teacher; small networks ([4, 3, 1] and [10, 6, 1]). Deeper or wider models, multiplication nodes and grid updates were not tested, so whether the same small μ suffices at scale is open.
- Two edge types, Gaussian RBFs and cubic B-splines on fixed grids, each with a SiLU base.
- For Gaussian-RBF edges our P1 is a surrogate of Bagrow's penalty (second differences instead of his curvature Gram matrix); his B-spline form was tested exactly (Section 8). We did not vary the SiLU term's weight.
- Receding behaviour is shown over $\gamma = 10^{-2}$ to 10^{-7} only, at $\rho = 0.95$ and on the real data; the dose thresholds are synthetic; and the strong real-data degradation from polishing did not appear in the synthetic study, so we do not claim it as a general law.
- The null-space weight in the shared-curve model was carried over from the synthetic calibration.

Post-hoc elements. The analyses listed in Appendix B were defined after seeing results; we report them as exploratory, never in place of pre-registered verdicts, and say where their rules were fixed before computing.

Open questions. A theory of the tilted valley (its curvature, its rate of recession, and when a distant finite minimum exists instead) is open. So are combining the null-space penalty with a concurvity regulariser (Siems et al., 2023), quantifying how much of the residual seed disagreement at certified minima matters scientifically, and checking whether runaway edges violate known electrolyte physics.

Process. Each study committed its predictions, thresholds and interpretation rules before its code existed, and every verdict is linked to its commit; following up failed predictions, rather than reinterpreting them, led to the mechanism reported here. Appendix D describes the protocol and the AI assistance used under it, and lists every change made after an outcome was known, including five to the B-spline study's calibration step, none of which changed a prediction, threshold or interpretation rule. A planned predictive follow-up was dropped because its precondition was not met.

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A All pre-registered predictions

Generated by `paper2/tex/build_tex.py` from the committed verdict sections; the cell text is the committed wording. Each study opens with its pre-registration's title and its committed question (or theory). Commits are given as pre-registration → results (in the anonymous version, as neutral labels S-NN). Thresholds are part of each prediction's wording.

A.1 Synthetic controlled study

Pre-registration: "Pre-registration: synthetic teacher-student test of the runaway mechanism (paper2/synth)". **Teacher and students: KAN [4, 3, 1].** Two stacked `kan.KANLayer`s with tanh between them, as in `kan.KAN`, but **without** the salt embedding that `kan.KAN` always adds. Same edge type: SiLU base plus 8 Gaussian RBFs on fixed centres linspace(−1, 1, 8), width 2/7.

ID	Prediction (threshold)	Result	Verdict	Commits
X1	runaway ($\geq$ 10) in correlated P0 and P1; $\leq$ 3 in the three independent cells and correlated P2	Correlated P0 ×65.7 and P1 ×32.1, and P2 ×1.00, as predicted. **But independent P0 ×12.5 and P1 ×7.9 exceed 3** (independent P2 ×1.09)	**failed**	dfd281d→ 34cc321
X2	certified $\leq$ 2/8 in correlated P0 and P1; $\geq$ 6/8 elsewhere	correlated P0 0/8, P1 0/8; independent 6/8, 6/8, 8/8; correlated P2 8/8	**held**	dfd281d→ 34cc321
X3	cancellation in correlated P0 and P1 each $\geq$ 2 × correlated P2	P0 3.29 and P1 2.32, against 2 × 1.99 = 3.98	**failed**	dfd281d→ 34cc321
X4	off-manifold ratio $\geq$ 3 in correlated P0 and P1, $\leq$ 1.5 in P2	P0 1.22, P1 **0.77**; P2 0.78 (the P2 part held)	**failed**	dfd281d→ 34cc321
X5	Spearman(cancellation, off-manifold RMSE) $\geq$ 0.6 over the 24 correlated models	ρ = +0.177, p = 0.407	**failed**	dfd281d→ 34cc321
X6	unit match $\geq$ 0.85 in the independent cells and correlated P2; < 0.7 in correlated P0 and P1	The first part held: 1.000 in all four no-runaway-predicted cells. **The second part failed: correlated P0 0.774 and P1 0.883 are not < 0.7**	**failed**	dfd281d→ 34cc321

A.2 Real-data case study

From the case-study close-out: KAN-δ predicts Li-electrolyte conductivity for the CALiSol-23 DEC holdout. Its prediction is log k = a measured pure-PC anchor + x_co · h(solvent descriptors, 1/T, molality, salt). Its first layer is a [8 continuous inputs + 2 salt-embedding, 6, 1] FastKAN, with SiLU + 8 Gaussian RBFs per edge and a smoothness penalty (`kan.py`, `final.py`).

Rows are as committed in the case-study summary (one row may cover several predictions, e.g. F2–F5). The rank intervention's verdicts are given per arm in the result column.

Study	Prediction (threshold)	Result	Verdict	Commits
Gauge study	**Control 1**: teacher edges recovered, median dense corr $\geq$ 0.8	0.301 (sanity: h corr 0.913 < 0.95)	**failed**	c11dd92→ e68e2eb
Gauge study	P1: gauge fixing removes $\geq$ 50% of edge variance	not run	inconclusive (gated by Control 1)	c11dd92→ e68e2eb
Gauge study	P2: sparse $\geq$ 2× dense, DEC region highest	not run	inconclusive (gated)	c11dd92→ e68e2eb
Gauge study	P3: (e) < (d) < (c) < (a)	not run	inconclusive (gated)	c11dd92→ e68e2eb
Gauge study	P4: every seed has a unit with |corr| < 0.8	not run	inconclusive (gated)	c11dd92→ e68e2eb
Gauge study	P5: KAN-full edge variance > KAN-δ	not run	inconclusive (gated)	c11dd92→ e68e2eb
Gauge study	P6: seed spread flags errors	Spearman +0.169 (A) / +0.213 (B); fold B RMSE at 80% coverage 0.153 vs random 0.137	**failed**	c11dd92→ 67fe9f2
Control 2	C1: (I,C) edges $\geq$ 0.9 and units $\geq$ 0.9	prediction corr 0.947 < 0.99	inconclusive (gate failed)	45b336f→ a260aef
Control 2	C2: (R,C) invT, molal $\geq$ 0.9; co-solvent < 0.7	prediction corr 0.969 < 0.99	inconclusive (gate failed)	45b336f→ a260aef
Control 2	C3: 107 epochs gives prediction corr < 0.95	0.858 (I,S), 0.919 (R,S)	**held**	45b336f→ a260aef
Control 3	C1': (I) passes the gate, edges $\geq$ 0.9, units $\geq$ 0.9	(I) RMSE/std 0.128 > 0.05; unit match 0.648	**failed**	812a0fa→ 9f59d39

Study	Prediction (threshold)	Result	Verdict	Commits
Control 3	C2': (R) edges $\leq$ (I) edges $-$ 0.10	(R) edges 0.852, units 0.659	inconclusive ((I) failed the gate)	812a0fa→ 9f59d39
Control 3	C4: every continuous input $\geq$ 0.9 in (I)	–	inconclusive ((I) failed the gate)	812a0fa→ 9f59d39
Fiber	F1: $\geq$ 50% of seed pairs on-fiber	0/190 (closest d = 0.146)	**failed**	e83df2e→ f220a2f
Fiber	F2–F5	not run	inconclusive (no on-fiber pairs)	e83df2e→ f220a2f
Fiber 2	G1: $\geq$ 4/6 seeds and $\geq$ 10/20 students certify convergence	0/6, 0/20 (all at the 20,000-iteration cap)	**failed**	4af5250→ 98eca9d
Fiber 2	G2–G5	not run	inconclusive (no converged model)	4af5250→ 98eca9d
Support	H1: dim(N) $\geq$ 5	dim(N) = 36 of 72	**held**	b473232→ c5505e5
Support	H2: $\geq$ 50% of N is REDISTRIBUTION within the correlated group	18/36	**held, exactly at threshold**	b473232→ c5505e5
Support	H3: set A fraction in N $\geq$ 0.5 and $\geq 3\times$ random	median 0.942 (random 0.5)	inconclusive (untestable: $3 \times 0.5 > 1$)	b473232→ c5505e5
Support	H4: set B (capped), same criterion	median 0.949	inconclusive (untestable)	b473232→ c5505e5
Support	H5: flattest Hessian directions overlap N $\geq$ 0.5	median 0.885; median layer-1 block share only 0.08	**held**, with caveat (see `support_prereg.md`)	b473232→ c5505e5
Support	H6: fraction in N rises from control-3 to capped state, $\geq$ 15/20	20/20 (0.713 $\rightarrow$ 0.949)	**held**	b473232→ c5505e5
Rule	I1–I5 (gate)	0 columns removed; gate stopped	inconclusive	6c93bf4→ a53cc2a
Rank	I1: median max|grad| $\leq$ 0.1 $\times$ baseline	arm G: **failed** (4.00×); arm F: **failed** (29.1×)	see result	df44ead→ aeb6f87
Rank	I2: median d_AB $\leq$ 0.5 $\times$ baseline	arm G: **failed** (0.945×); arm F: **failed** (0.808×)	see result	df44ead→ aeb6f87
Rank	I3 (sanity): median loss $\leq$ 1.10 $\times$ baseline	arm G: passed (1.004×); arm F: passed, narrowly (1.099×)	see result	df44ead→ aeb6f87
Rank	I4: fold A RMSE $\leq$ 0.150 (KAN-δ 0.1420)	arm G: **held**, narrowly (0.1499); arm F: **failed** (0.1587)	see result	df44ead→ aeb6f87
Rank	I5: edge corr on 1000/T and molality $\geq$ baseline + 0.1 (baseline 0.102)	arm G: **held** (0.927); arm F: **held** (0.338)	see result	df44ead→ aeb6f87
Rank	I6: arm F's improvement $\geq$ arm G's in both I1 and I2	arm G: **failed** (max|grad| 29.1× vs 4.00×; d_AB does favour F, 0.808 vs 0.945)	see result	df44ead→ aeb6f87
Trust	T1 / T2: trust-krylov certifies $\geq$ 4/6 seeds / $\geq$ 2/3 students	0/6, 0/3. All 9 stopped on scipy's "bad approximation" after 1–62 iterations; the Hessian was accurate to $\leq 2 \times 10^{-7}$	**failed**	a77e8b4→ 67586d6
Trust	T3–T5	nothing converged	inconclusive	a77e8b4→ 67586d6
Stage 2	U1: trust-exact (exact Hessian, 500 iterations) certifies $\geq$ 4/6 seeds	0/6. All 9 hit the iteration cap	**failed**	c6a9e83→ b8d6488
Stage 2	U2: with L2 $\lambda = 10^{-5}$, $\geq$ 4/6 seeds and $\geq$ 2/3 students	1/6 seeds, 0/3 students	**failed**	c6a9e83→ b8d6488
Stage 2	U3: more converge at 10^{-5} than at 10^{-6}	1 vs 0	**held**	c6a9e83→ b8d6488
Stage 2	U4: converged L2 models' loss without L2 $\leq$ 1.05 $\times$ capped	1.2545	**failed**	c6a9e83→ b8d6488
Stage 2	U5: converged seeds' disagreement $\leq$ 0.5 $\times$ capped	only 1 converged	inconclusive	c6a9e83→ b8d6488

Study	Prediction (threshold)	Result	Verdict	Commits
Close	V1: at some μ, $\geq 4/6$ seeds and $\geq 2/3$ students converge (stage2 criterion)	0 certified at every μ	**failed**	e5105d7→ 65ed55f
Close	V2: converged seeds' data term $\leq 1.05 \times$ stage2 A	nothing certified at the selected $\mu = 10^{-6}$	inconclusive	e5105d7→ 65ed55f
Close	V3: $\|\theta\|^2 \leq 10 \times$ start	same	inconclusive	e5105d7→ 65ed55f
Close	V4: seed disagreement halved	same	inconclusive	e5105d7→ 65ed55f
Close	V5: students' unit match ≥ 0.8	same	inconclusive	e5105d7→ 65ed55f
Close	V6: fold A RMSE ≤ 0.150	1.0262, bias +0.9028	**failed**	e5105d7→ 65ed55f
Certify	W1: at $\mu = 10^{-4}$, $\geq 5/6$ seeds certified (second-order criterion, F-a)	6/6; all 17 candidates certified	**held**	5f4520a→ 36f295b
Certify	W2: certified seeds' median d_AB $\leq 0.5 \times$ capped	0.0771 vs capped 0.1041 (-26%)	**failed**	5f4520a→ 36f295b
Certify	W3: certified students' unit match ≥ 0.8 (vs the original teacher)	0.624	**failed**	5f4520a→ 36f295b
Certify	W4: $\mu = 10^{-4}$ fold A RMSE $\leq$ 0.150	0.3345	**failed**	5f4520a→ 36f295b
Certify	W5: $\mu = 10^{-4}$ fold A RMSE $\leq$ 0.5131	0.3345	**held**	5f4520a→ 36f295b
Certify	W6: Spearman(cross-edge ratio, held-out RMSE) ≥ 0.6	$\rho = +0.345$, p = 0.328 (weak test)	**failed**	5f4520a→ 36f295b

Committed tally of the case study:

Study	Held	Failed	Inconclusive
First table (Control 1, P1–P6, C1–C3, C1'/C2'/C4, F1–F5, G1–G5, H1–H6)	5	5	19
Rule (I1–I5)	0	0	5
Rank (I1, I2, I4, I5 per arm, I6)	3	6	0
Trust (T1–T5)	0	2	3
Stage 2 (U1–U5)	1	3	1
Close (V1–V6)	0	2	4
Certify (W1–W6)	2	4	0
Total	**11**	**22**	**32**

A.3 Shared-curve KAN

Pre-registration: "SC-KAN-δ (shared-curve KAN-δ): pre-registration". **Inputs:** the 8 continuous final.py features (flag 11), scaled exactly as in final.py. Salt enters through a learned 2-number embedding e(salt), passed through tanh.

Run after the case-study close-out; not part of the 65 predictions above.

ID	Prediction (threshold)	Result	Verdict	Commits
S1	SC LOCO RMSE $\leq 1.05 \times$ baseline	unweighted mean SC 0.2923 vs base 0.2554 (ratio 1.145; threshold 0.2681). Also reported: row-weighted 0.2478 / 0.2297 (1.079); folds ≥ 20 rows 0.3138 / 0.2093 (1.499)	failed	0053d01→ e00e37f
S2	SC fold A RMSE ≤ 0.160	SC 0.1757 (base 0.1420)	failed	0053d01→ e00e37f
S3	SC median d_AB $\leq 0.5 \times$ baseline	SC 0.1865 vs base 0.1120 (threshold 0.0560): SC seeds agree *less* than baseline seeds	failed	0053d01→ e00e37f

ID	Prediction (threshold)	Result	Verdict	Commits
S4	curve corr ≥ 0.90 for ≥ 6 of 8 inputs	3 of 8 (M_co 0.959, mix_lneta 0.970, invT 0.986); lowest x_co 0.596; others 0.871–0.891	failed	0053d01→ e00e37f
S5	$\geq 8/10$ SC fold A models certified	SC 10/10 (max|grad| $\leq 7 \times 10^{-10}$, 0 near-null eigenvalues); baseline 0/10 (all hit the 500-iteration cap)	held	0053d01→ e00e37f
S6	median β-norm polished/early ≤ 3 (calibrated)	median 1.337 (range 0.652–2.356); $\|\theta\|^2$ $\times 4.59$ (base $\times 67.9$, range 30.6–281)	held	0053d01→ e00e37f
S7	SC polished/early fold A RMSE $\leq$ 1.5	SC 1.920 (0.1757 → 0.3373); baseline 5.736 (0.1420 → 0.8147)	failed	0053d01→ e00e37f
S8	d log k/d(1000/T) < 0 on $\geq 95\%$ rows in $\geq 8/10$ seeds	10/10 (median share 1.000; 1424 rows, 10 excluded per D4)	held	0053d01→ e00e37f
S9	Spearman(1000/T, |slope|) > 0 in $\geq 8/10$ seeds	10/10 (median +0.661, range +0.406 to +0.885)	held	0053d01→ e00e37f
S10	d log k/d(ln η_mix) < 0 on $\geq 90\%$ rows in $\geq 8/10$ seeds	7/10 (median share 0.967; s0 0.892, s2 0.892, s8 0.826)	failed	0053d01→ e00e37f

A.4 Two-anchor excess model

Pre-registration: "EX-KAN (two-anchor excess model): pre-registration". log k = anchor_PC + x_co · s(eps_co, ln_eta_co, M_co, 1000/T) + x_co · (1 − x_co) · g(eps_co, ln_eta_co, M_co, 1000/T, molality, salt) So δ = log k − anchor_PC = x_co · [s + (1 − x_co) · g].

Run after the case-study close-out; not part of the 65 predictions above.

ID	Prediction (threshold)	Result	Verdict	Commits
E1	EX LOCO $\leq 0.95 \times$ KAN-δ	unweighted mean EX 0.2834 vs threshold 0.2426 (base 0.2554; ratio 1.110). Row-weighted 0.2363 / 0.2297 (1.029); folds ≥ 20 rows 0.2662 / 0.2093 (1.272). EX is better on EC (0.237 vs 0.256), Sulfolane, 3-Glyme, 2-Glyme and Toluene, and worse on 2-MeTHF, THF, Methylene chloride, DME and EA	failed	3938f8f→ 579b58f
E2	EX fold A ≤ 0.142	EX 0.1502 (base this run 0.1420)	failed	3938f8f→ 579b58f
E3	s agrees across seeds (separable co-solvents, Amendment 1)	median pairwise corr +0.987 over EA, EC, Methylene chloride: a weak test with 3 values per seed, and Methylene chloride is marginal. Seed SD of s: EA 0.063 (0.39 × the spread across co-solvents, 0.159), EC 0.091 (0.57 ×), Methylene chloride 0.067 (0.42 ×). Secondary (≤ 2 rule, 7 co-solvents): +0.869	held	3938f8f→ 579b58f
E4	mean s(low ε) $<$ mean s(EC, Sulfolane) in $\geq 8/10$ seeds	0/10: the ordering is reversed in every seed. Per high-ε co-solvent: vs EC alone 0/10 (EC median s +0.104), vs Sulfolane alone 0/10 (median −0.124). The low-ε mean is +0.19 to +0.49 per seed. Sulfolane does not drive the failure on its own: EC alone fails in every seed too	failed	3938f8f→ 579b58f
E5	median seed-SD flagged $\geq 2 \times$ separable (Amendment 1)	0.091 vs 0.067, ratio 1.37. Secondary (≤ 2 rule): 0.222 vs 0.069, ratio 3.22, no verdict; the secondary is driven by 3-Glyme (SD 0.478, M_co extrapolated) and 2-Glyme (0.222)	failed	3938f8f→ 579b58f

Revision studies. The studies below were run for the revision, each with its own pre-registration.

A.5 Freeze test: is composition necessary? (and weight decay)

Pre-registration: "Revision step 2: is composition necessary? (plus weight decay)". With layer 1 fixed, the output is linear in the layer-2 parameters: a fixed basis times coefficients. The data term plus a quadratic penalty is then a convex quadratic. It can have flat directions but no tilted ones, so it cannot run away. If the runaway needs composition, freezing layer 1 should remove it.

ID	Prediction (threshold)	Result	Verdict	Commits
Z1	Arm A certified: $\geq 5/6$ real; $\geq 7/8$ in each synthetic cell	A-R 6/6; A-S correlated P0 8/8; A-S correlated P1 8/8	HELD	a6d752b→ 9228f70
Z2	Arm A median layer-2 growth ≤ 3 in R and in each synthetic cell	A-R ×582.8 (×206–×11,689); P0 ×4.07 (×1.36–×245); P1 ×22.9 (×1.51–×801)	FAILED	a6d752b→ 9228f70
Z3	Arm A smallest Hessian eigenvalue $\geq -10^{-8} \times$ largest at every logged iterate	worst ratio $+1.99 \times 10^{-10}$ (all values positive)	HELD	a6d752b→ 9228f70
Z4	Arm C: 8/8 certified, median growth ≤ 3	8/8; ×2.52 (×1.68–×2.85)	HELD (computed before the stop)	a6d752b→ 9228f70
Z5	Arm D branch	8/8 certified; median growth ×0.415 (×0.263–×0.540). Note: this rule uses the same ≤ 3 growth threshold as Z2 and shares its weakness (see the interpretation note); here the norm shrinks, so the threshold is not binding.	"an L2 term of this size at polish time bounds the runaway"	a6d752b→ 190544d
Z6 E(i)	Arm E, synthetic correlated P0	median $\|\theta\|^2$ ratio wd 0 / wd 10^{-4} ×1.0011 (×0.987–×1.007); in-distribution RMSE 0.0091 vs 0.0091; off-manifold 0.0667 vs 0.0665; determinism check max |diff| 0	"early stopping alone kept them bounded; weight decay was incidental"	a6d752b→ 190544d
Z6 E(ii)	Arm E, real fold A	median ratio ×1.0001 (×1.0001–×1.0001); fold A ensemble RMSE 0.1420 vs 0.1420; determinism check max |diff| 0	"early stopping alone kept them bounded; weight decay was incidental"	a6d752b→ 190544d

A.6 Infimum test: a certified regularisation path

Pre-registration: "Revision step 3: infimum test (a certified L2 regularisation path)". Does the unregularised loss of the full model have a finite minimum that is merely distant? Or does its infimum lie at infinity (a receding minimum)?

Predictions I1–I4 apply per phase (Phase 1 synthetic, Phase 2 real data); the real-data control arm was added after Phase 2 with its outcomes fixed before it ran.

ID	Prediction (threshold)	Result	Verdict	Commits
I1	Phase 1: runaway: $\|\theta^*\|^2$ increases as γ decreases	16/16	HELD	179814f→ ef3b3c2
I2	Phase 1: runaway: $\alpha > 0.2$, bootstrap 95% interval excluding 0	16/16	HELD	179814f→ ef3b3c2
I3	Phase 1: control: ratio between the two smallest certified $\gamma \leq 1.5$	8/8	HELD	179814f→ ef3b3c2
I4	Phase 1: runaway: L^* decreases as γ decreases	16/16	HELD	179814f→ ef3b3c2
I1	Phase 2: runaway: $\|\theta^*\|^2$ increases as γ decreases	6/6	HELD	179814f→ 349022b

ID	Prediction (threshold)	Result	Verdict	Commits
I2	Phase 2: runaway: $\alpha > 0.2$, bootstrap 95% interval excluding 0	6/6	HELD	179814f→ 349022b
I3	Phase 2: discrimination check	—	Phase 1's I3 (HELD, 8/8), as pre-registered; Phase 2 has no control models	179814f→ 349022b
I4	Phase 2: runaway: L^* decreases as γ decreases	6/6	HELD	179814f→ 349022b
Control	Post hoc, outcomes fixed before running. Plateau: the $\|\theta^*\|^2$ ratio between the two smallest certified (incl. certified-degenerate) γ values is ≤ 1.5 for both seeds. The real-data contrast is then complete.	The $\|\theta^*\|^2$ ratio between the two smallest certified γ is 1.092 (ind0) and 1.101 (ind1), both ≤ 1.5, so the real-data contrast is complete. The divergence condition is not met: α is +0.105 [+0.093, +0.118] and +0.124 [+0.109, +0.139], both below 0.2.	plateau	349022b→ 4003b21

A.7 Correlation dose-response

Pre-registration: "Revision step 4: dose-response — how much correlation does the blind spot need?". At what input correlation does the runaway turn on, and does the null-space penalty still close it at the strongest dose?

Verdict column: P0 / P1. The thresholds are synthetic ([4, 3, 1], Gaussian inputs), not real-data thresholds.

ID	Prediction (threshold)	Result	Verdict	Commits
R1	The uncertified fraction (measure 1) is non-decreasing across ρ, for P0 and for P1 separately, allowing one inversion.	P0: HELD: 0.125, 0.625, 0.500, 1, 1, 1 (1 inversion); P1: HELD: 0.125, 0.500, 0.500, 0.625, 1, 1 (0 inversions)	held / held	1a18569→ 541355a
R2	Control, P2 at $\rho = 0.99$: at least 7 of 8 certified (including certified-degenerate), and median D ≤ 1.5.	P2 at $\rho = 0.99$: 8/8 certified, median D 1.127 (range of growth ×0.88–×1.11).	held	1a18569→ 541355a
R3	The median affine-RBF share of the polished change is non-decreasing in ρ, for P0 and for P1 (one inversion allowed).	P0: FAILED: 0.715, 0.594, 0.612, 0.476, 0.394, 0.585 (3 inversions); P1: FAILED: 0.944, 0.982, 0.748, 0.934, 0.914, 0.952 (2 inversions)	failed / failed	1a18569→ 541355a
R4	Median D increases with ρ, for P0 and for P1: Spearman correlation ≥ 0.8 between ρ and the cell median D over the 6 levels. (Spearman ≥ 0.8 is the operative criterion; it tolerates up to about two inversions.)	P0: FAILED: D 8.01, 15.05, 6.43, 9.52, 16.13, 16.72; Spearman +0.714; P1: FAILED: D 4.54, 4.88, 2.91, 3.00, 4.52, 14.54; Spearman +0.143	failed / failed	1a18569→ 541355a
	R4, certified-only variant	P0: FAILED (Spearman +0.714; D at 0.99 is 19.55, n = 4); P1: FAILED (Spearman +0.143)	failed / failed	1a18569→ 541355a

A.8 B-spline replication

Pre-registration: "Revision step 5: B-spline replication of the synthetic 2 × 3 study". Are the runaway and its fix a property of KAN-style edges with second-difference penalties, or an artefact of the Gaussian-RBF basis?

Step 1 verdicts at the frozen $\mu = 10^{-6}$, then the post-hoc follow-up checks whose predictions and readings were fixed before computing (check 3 is descriptive; Appendix B).

ID	Prediction (threshold)	Result	Verdict	Commits
B1	Correlated P0 and correlated P1: at most 2 of 8 certified, each.	correlated P0: 0/8 (8 iteration cap)	HELD	5491481→ f0accc0
		correlated P1: 0/8 (8 iteration cap)	HELD	5491481→ f0accc0
B2	P2, both input conditions: at least 7 of 8 certified AND median D $\leq$ 1.5, each.	independent P2: 8/8, D 1.222 (certified-only 1.222)	HELD	5491481→ f0accc0
		correlated P2: 8/8, D **2.225** (certified-only 2.225)	FAILED (the D leg; certification held)	5491481→ f0accc0
B3	Independent P0 and P1: at least 6 of 8 certified, each (the Gaussian reference level).	independent P0: 5/8 (3 iteration cap)	FAILED	5491481→ f0accc0
		independent P1: 4/8 (3 iteration cap, 1 solver error via the watchdog)	FAILED	5491481→ f0accc0
Check 1	Post hoc, predictions fixed in advance; P2 path extended to $\gamma = 10^{-9}$: median D' $\leq$ 1.5 in both conditions;; median norm ratio $\|\theta^*(10^{-9})\|$ / $\|\theta_\text{polish}\|$ within [0.9, 1.1] in both conditions.	independent: D' 1.003, norm ratio 1.000; correlated: D' 1.035, norm ratio 0.911	both hold	ff5bd20→ 9167740
Check 2	Post hoc, prediction fixed in advance; P2 at $\mu = 10^{-5}$: $\geq$ 7/8 certified and median D $\leq$ 1.5 in both conditions.	independent: 8/8 certified, median D 1.167 (1.167); correlated: 8/8 certified, median D 1.309 (1.309)	held / held	ff5bd20→ 9167740

B Post-hoc analyses

Every analysis below except one was defined after seeing results; the exception, the applicability analysis, is descriptive, with its plan committed before computing. Each row's label states how the analysis was defined, as recorded in its own files; none changes a pre-registered verdict, and all are reported as exploratory. The description is the subject line of the commit that added the analysis (plan and results commits where both exist).

Analysis	Commit subject (as committed)	Commit
Post hoc. Anatomy of the real-data runaway `anatomy_plan.md, anatomy.py, anatomy_output.txt`	Plan for descriptive runaway anatomy (before anatomy.py exists)	256b00a
	Anatomy (descriptive): Q1, Q2 not as expected, Q3 as expected; RBF growth is in the penalty null space	a791652
Post hoc. Gauge check and regularised teacher (exploratory) `gaugecheck.py, gaugecheck_notes.md, gaugecheck_output.txt`	gaugecheck (POST-HOC, exploratory): penalty fixes gauge position by slope balance; students match the regularised teacher	0b29351
Post hoc. Anatomy of the synthetic runaway `paper2/anatomy_synth.py, paper2/anatomy_synth_output.txt`	paper2 anatomy_synth (POST-HOC, descriptive): synthetic runaway lives mainly in layer-2 output-edge cancellation, unsaturated; no verdicts change	ea22a56
Post hoc. Loss decomposition along the runaway `paper2/lossdecomp.py, paper2/lossdecomp_output.txt`	paper2 lossdecomp (POST-HOC, descriptive): synthetic runaway loss decrease shared, D2 penalty median 0.61 vs data 0.37; 12/20 runaway models overfit; real-data reference recomputed	4ec6fce
Post hoc. Descriptive checks (hidden-unit correlation, tilt affinity, references) `paper2/checks/plan.md, paper2/checks/checks.py, paper2/checks/checks_output.txt`	paper2/checks: plan for descriptive checks (F4 hidden-unit correlation, F2 tilt affinity, A1 commit and data citation) before checks.py exists	5745103

Analysis	Commit subject (as committed)	Commit
	paper2/checks (POST-HOC, descriptive): hidden-unit correlation marks the runaway (independent cells 0.70 vs 0.05) but is not sufficient; RBF tilt affine to 2% centrally, 13% at the ends; A1 commit 3081cc0; CALiSol-23 citation not in competition material	6e1df32
Post hoc. Degeneracy follow-up: restart, scaling and flat-direction probes of the freeze test's endpoints `paper2/revision/degeneracy.py, paper2/revision/degeneracy_output.txt`	Post-hoc degeneracy follow-up: crash record 2 (trust-exact subproblem hang in the B-R ind0 restart probe; timer did not fire) and implementation-only fix (restart probes in a killable child process with checkpoints)	dc07ec5
Post hoc. Infimum test, Phase 0: re-classification of the freeze test's degenerate endpoints `paper2/revision/infimum_prereg.md`	Infimum test Phase 0 (POST-HOC, no class changed) and Phase 1 (I1-I4 held 16/16, 16/16, 8/8, 16/16: receding minimum over gamma 10^{-2}..1e-7); post-hoc cold-start basin check (no small-norm basin; better larger-norm minima at small gamma); verdicts appended	ef3b3c2
Post hoc. Infimum test: cold-start basin check `paper2/revision/infimum_prereg.md`	Infimum test Phase 0 (POST-HOC, no class changed) and Phase 1 (I1-I4 held 16/16, 16/16, 8/8, 16/16: receding minimum over gamma 10^{-2}..1e-7); post-hoc cold-start basin check (no small-norm basin; better larger-norm minima at small gamma); verdicts appended	ef3b3c2
Post hoc, outcomes fixed in advance. Infimum test: real-data control arm (null-space penalty at $\mu = 10^{-4}$) `paper2/revision/infimum_prereg.md`	Infimum test: Phase 2 verdicts; pre-register POST-HOC real-data control arm (before it runs)	349022b
	Infimum test Phase 2 (I1, I2, I4 held 6/6; receding minimum over gamma 10^{-2}..1e-7; six distinct branches), arms (a)/(c) (descriptive, n = 3), POST-HOC real-data control arm (mu = 10^{-4} fixed: PLATEAU, ratios 1.09/1.10), cold-start summary; final run notes (interruption = session end, cumulative cap held; Phase 1 equivalence hashes)	4003b21
Post hoc, rule fixed in advance. Dose-response settling check (short path against the certified polished minimum) `paper2/revision/dose_posthoc.py, paper2/revision/dose_posthoc_results.json`	dose_posthoc.py: POST-HOC descriptive check (short path vs certified polished minimum), rules fixed before computing	360a79c
	Dose-response POST-HOC check: 30/32 certified-polish models read FINITE (11 valley, 19 not yet reached), 0 RECEDING, 2 unclassified -> reading A below Branch 5; run notes a-d	75b778b
Post-outcome diagnostic, criteria fixed in advance. B-spline null-space diagnostic (constant-shift symmetry) `paper2/revision/bspline_nullspace_diag.py, paper2/revision/bspline_nullspace_diag.json`	bspline_nullspace_diag.py: DESCRIPTIVE constant-shift null-space diagnostic (criteria fixed before results)	baaa898
	B-spline null-space diagnostic (DESCRIPTIVE): exact 15-dim constant-shift symmetry confirmed (exactness, containment, RBF contrast, hangs after convergence); criterion (ii) fails at the two correlated end points (16 and 3 extra near-flat, non-exact directions at 10^{-10}..1e-8); stop	e2a2160
Post hoc, predictions fixed in advance. B-spline follow-up: path extension, $\mu = 10^{-5}$, the non-certifying independent models `paper2/revision/bspline_prereg.md, paper2/revision/bspline_posthoc.py, paper2/revision/bspline_posthoc_results.json`	B-spline prereg: POST-HOC addendum (P2 path extension, P2 at mu = 10^{-5}, non-certifying independent models), predictions and readings fixed before computing	ff5bd20

Analysis	Commit subject (as committed)	Commit
	B-spline POST-HOC follow-up: check 1 both predictions hold (D' 1.003 / 1.035; norm ratio 1.000 / 0.911), check 2 holds at mu = 10^{-5} (8/8, D 1.167 / 1.309), check 3 2 runaway-like / 4 mixed / 1 slow; archive 18	9167740
Descriptive, plan committed first. Applicability analysis (no predictions) `paper2/revision/appdomain_plan.md`, `paper2/revision/appdomain.py`, `paper2/revision/appdomain_output.txt`	Applicability-domain plan (DESCRIPTIVE; definitions and figures fixed before computing; facts of record; G1-G4)	ec769ad
	Applicability domain (DESCRIPTIVE): fold ensembles reproduced (0.1420 / 1.0262 / 0.3345); off-pattern tracks LOCO RMSE better than general distance but n = 7, not significant; within fold A polishing hurt most near the training data, collinear with x_co; fold A on-pattern, DEC 7x further off-pattern; parked study's precondition not met	656b06c
Post hoc. Applicability: artefact check (the ensembles' own errors against distance) `paper2/revision/appdomain_results.json`	Applicability domain: close-out; parked predictive study dropped (precondition not met)	b19fa2f
Descriptive, no plan. Uncertainty: Wilson intervals, fold A composition bootstrap, stop reasons `paper2/revision/uncertainty.py`, `paper2/revision/uncertainty_output.txt`	uncertainty.py (DESCRIPTIVE, from committed results): Wilson intervals, fold A composition bootstrap, stop reasons	b1605d8
Post hoc, plan committed first. Edge decomposition of the fold A ensembles `paper2/revision/edgedecomp_plan.md`, `paper2/revision/edgedecomp.py`, `paper2/revision/edgedecomp_output.txt`	Edge-decomposition plan (POST-HOC, DESCRIPTIVE), committed before edgedecomp.py exists	207b22f
	Edge decomposition of the fold A ensembles (POST-HOC, DESCRIPTIVE; plan 207b22f)	fa5d961
Post hoc. Regularised teacher (weights pinned)	Pin gaugecheck regularised-teacher weights in a seventh archive; append to weights manifest	47d7327

C Reproducibility map

Generated from `paper2/figures/MANIFEST.md`. Every figure is rebuilt by its script from committed results; each figure's sidecar JSON lists the data files (path, commit, SHA-256), any archived weights (SHA-256, archive) and every plotted number. Code, committed results and figure scripts: `https://github.com/drgelavizhahmadi-dotcom/battery_hacketon`.

Figure	Script	Data files (commit)	Archived weights
Figure 1 (`fig1_mechanism`)	`make_fig1.py`	none (computed from committed code: kan.py, paper2/synth.py)	—
Figure 5 (`fig2_synthetic`)	`make_fig2.py`	`paper2/synth_results.json` (34cc321)	—
Figure 6 (`fig3_where`)	`make_fig3.py`	`weights_manifest.txt` (179ac24); `paper2/anatomy_synth_results.json` (ea22a56); `paper2/checks/checks_results.json` (6e1df32)	18 files, archives gauge_weights_2026-09-29.tar.gz, gauge_weights_2026-09-29b.tar.gz, gauge_weights_2026-10-02e.tar.gz

Figure	Script	Data files (commit)	Archived weights
Figure 7 (`fig4_applicability`)	`make_fig4.py`	`paper2/revision/appdomain_results.json` (b19fa2f); `weights_manifest.txt` (179ac24)	15 files, archives gauge_weights_2026-10-03f.tar.gz, gauge_weights_2026-10-04i.tar.gz, gauge_weights_2026-10-25s.tar.gz
Figure 2 (`fig5_paths`)	`make_fig5.py`	`paper2/revision/infimum_results.json` (4003b21)	—
Figure 3 (`fig6_correlation_dose`)	`make_fig6.py`	`paper2/revision/dose_results.json` (541355a)	—
Figure 4 (`fig7_bspline`)	`make_fig7.py`	`paper2/synth_results.json` (34cc321); `paper2/revision/bspline_results.json` (ffa6e64); `paper2/revision/bspline_posthoc_results.json` (9167740)	—
Figure 8 (`fig8_dose_response`)	`make_fig8.py`	`stage2_results.json` (b8d6488); `close_results.json` (65ed55f); `certify_results.json` (36f295b)	—
Figure 9 (`fig9_uniqueness`)	`make_fig9.py`	`arch/sc_results.json` (e00e37f); `certify_results.json` (36f295b); `weights_manifest.txt` (179ac24)	40 files, archives gauge_weights_2026-10-04i.tar.gz
Figure 10 (`figS1_lossdecomp`)	`make_supp.py`	`paper2/lossdecomp_results.json` (4ec6fce)	—
Figure 11 (`figS2_gaugecheck`)	`make_supp.py`	`gaugecheck_results.json` (0b29351)	—
Figure 12 (`figS3_loco`)	`make_supp.py`	`arch/sc_results.json` (e00e37f); `arch/ex_results.json` (579b58f)	—

Tables in the main text (no script; values typed in the draft):

Table	Source	Data files	Commits
Table 1	typed in the draft from	`paper2/synth_results.json`, `paper2/synth_output.txt`	34cc321
Table 2	typed in the draft from	`close_results.json`, `certify_results.json`, `stage2_results.json`	65ed55f, 36f295b, b8d6488
Table F.1	typed in the draft from	`arch/sc_results.json`, `arch/ex_results.json`, `paper2/figures/fig9_uniqueness.json`	e00e37f, 579b58f, 9c38c29

Panels evaluated post hoc from archived weights (from `MANIFEST.md`):

- **fig3a, real data:** the split of the parameter change into affine and curved RBF parts per layer, for 107 epochs → fiber2 capped → stage2 Part A.
- `stage2_results.json` stores only per-block norms, without an affine/curved split and without the 107 → capped change.
- **fig9b, polished fold A d_AB** for SC-KAN-δ and the baseline (marked *).
- arch/sc committed d_AB only at the early state.
- **fig4a, b, fold A per-row error change** (early-stopped -> mu = 10^{-6} and mu = 10^{-4} ensembles).
- appdomain_results.json stores only the per-row Spearman correlations and medians; the per-row values are re-evaluated with the

D Research process

Pre-registration. Every study committed a pre-registration file (predictions, thresholds, interpretation rules) before the script that tested it existed. Verdicts were appended below the committed text; the pre-registered text was never edited, and corrections to reported numbers were made in separate, labelled commits.

Amendments. A change to a design after its pre-registration was committed as a dated amendment section before any affected model was trained, with its reason; the original text was kept. Choices forced by the code or data at run time were logged as numbered discrepancies at the top of each output log.

Files win. Instructions to the agent stated that committed files take precedence over any number or description in a prompt; discrepancies between a brief and the files were flagged rather than resolved silently.

Data and weights. Model weights were never committed: they were archived outside the repository with a manifest of SHA-256 hashes and producing commits, and every later analysis verified the hashes and held its inputs in memory. Post-hoc analyses were labelled as such in their files and changed no verdict.

AI assistance. Code, runs and logs were produced by an AI coding agent working under this protocol. Study designs, briefs and the manuscript were developed with an AI assistant. The author set the research questions, reviewed and approved every brief and amendment, decided each next stage, and takes responsibility for all content. Crashes were recorded in the logs and fixed in separate commits labelled as implementation-only.

Pre-registrations, amendments and crash records (from the commit history):

Commit	Date	Kind	Commit subject
c11dd92	2026-09-23	pre-registration	Pre-register KAN identifiability study (before any experiment)
45b336f	2026-09-23	pre-registration	Pre-register corrected 2x2 control (before writing or running it)
812a0fa	2026-09-23	pre-registration	Pre-register control 3: reachable teacher (before writing code)
e83df2e	2026-09-29	pre-registration	Pre-register fiber completeness test (before writing or running fiber.py)
4af5250	2026-09-29	pre-registration	Pre-register fiber2: certified convergence + teacher-known fiber analysis (before writing code)
b473232	2026-09-29	pre-registration	Pre-register support study: near-null space, where models differ, Hessian (before support.py exists)
175478a	2026-09-29	pre-registration	Close-out: electrolyte case study summary (all pre-registered predictions)
6c93bf4	2026-09-29	pre-registration	Pre-register basis-size rule intervention (before rule.py exists)
df44ead	2026-09-29	pre-registration	Pre-register rank-constrained layer-1 intervention (before rank.py exists)
a77e8b4	2026-09-30	pre-registration	Pre-register trust-region Newton polishing, stage 1 (before trust.py exists)
c6a9e83	2026-09-30	pre-registration	Pre-register stage 2: trust-exact check, runaway diagnostic, coercivity (before stage2.py exists)
a1c7842	2026-09-30	crash record / fix	stage2.py as first run (crashed with AttributeError in callback)
78bcaba	2026-09-30	crash record / fix	stage2: fix AttributeError (implementation only, post-prereg)
60621c6	2026-10-02	crash record / fix	stage2: hold verified inputs in memory (implementation only, post-prereg)
e5105d7	2026-10-02	pre-registration	Pre-register targeted null-space penalty intervention (before close.py exists)
5f4520a	2026-10-03	pre-registration	Pre-register second-order certification and regularised fold A accuracy (before certify.py exists)
dfd281d	2026-10-03	pre-registration	paper2: pre-register synthetic 2x3 runaway test (before synth.py exists)
0053d01	2026-10-03	pre-registration	arch: pre-register SC-KAN-δ (shared-curve KAN-δ) before any script exists; decisions C (centring), mu null-space term, tanh salt embedding, full beta term
27a8b33	2026-10-03	crash record / fix	arch sc_run: fix log() call (crash record 1, after the first fold A polish); resume from cache
3938f8f	2026-10-04	pre-registration	arch: pre-register EX-KAN (two-anchor excess model) before any script exists; decision B (x_co-independent g, [7,6,1]), full set for E3-E5, E4 mean rule
fa6ae2a	2026-10-04	amendment	arch ex_prereg: Amendment 1 (after Step 0, before training): primary separability rule by fixed-composition groups (EA, EC, Methylene chloride separable); committed rule reported as secondary
a6d752b	2026-10-16	pre-registration	Revision step 2: pre-register the composition test (frozen layer 1, one-layer control) and the weight-decay arms D/E, before freeze.py exists; draft rev 37
11bf8bc	2026-10-17	pre-registration	Revision step 2: decision record - freeze.py run stopped by the pre-registered rule (Z2 failed in Arm A); the author decided to resume (option 2)

Commit	Date	Kind	Commit subject
9228f70	2026-10-17	crash record / fix	Revision step 2: crash record 1 (trust-exact subproblem hang in Arm B-R) and implementation-only fix (timer-enforced 2 h cap); part-1 verdicts and post-hoc note in the prereg
dc07ec5	2026-10-18	crash record / fix	Post-hoc degeneracy follow-up: crash record 2 (trust-exact subproblem hang in the B-R ind0 restart probe; timer did not fire) and implementation-only fix (restart probes in a killable child process with checkpoints)
179814f	2026-10-18	pre-registration	Pre-register infimum test: certified L2 regularisation path (before infimum.py exists)
349022b	2026-10-19	pre-registration	Infimum test: Phase 2 verdicts; pre-register POST-HOC real-data control arm (before it runs)
1a18569	2026-10-20	pre-registration	Pre-register dose-response study (before dose.py exists)
d665db2	2026-10-20	crash record / fix	dose.py as it ran the replication gate (attempt 1, stopped by crash record 1)
1d53d85	2026-10-20	crash record / fix	dose.py: crash record 1 (SIGBUS in Accelerate dgemm during a polish Hessian) and implementation-only fix (a committed polish restarts from the early state; stale heartbeat discarded)
11e3663	2026-10-20	crash record / fix	dose/infimum: crash record 2 (non-finite trust-exact subproblem in scipy) and implementation-only fix (stop at the last accepted iterate with reason 'solver error')
5491481	2026-10-22	pre-registration	Pre-register B-spline replication (before bspline.py exists)
0ce6d00	2026-10-22	crash record / fix	B-spline basis (efficient-kan-style extended uniform knots, G = 5, k = 3) and property tests (a)-(e), implementation only
cba2c07	2026-10-22	amendment	B-spline prereg: amendment after Step 0 (i) failure; fidelity gate withdrawn as mis-specified (before any further run)
a2f750a	2026-10-22	crash record / fix	bspline: crash record 1 (Step 0 (ii) teacher polish hung in scipy's trust-exact subproblem) and implementation-only fix (teacher polishes in hard-killed children with checkpoints; ending classification per 4b3f20a; mu calibration as its own stage)
03638f8	2026-10-23	amendment	B-spline prereg: POST-HOC amendment after the null-space diagnostic (analytic 15-dim constant-shift null set; null-set-aware certification, degeneracy count and probes; certification-based termination; B1-B3 unchanged)
5fb9c47	2026-10-23	amendment	B-spline: null-set-aware certification, complement degeneracy count and probes, certification-based termination (amendment 03638f8); re-certification and validation stages; tests
2834b26	2026-10-23	crash record / fix	B-spline: staleness watchdog (kill a child with no new accepted iterate for 10 min; 'solver error (hang)'), implementation-only crash handling (note 5bf5acd); tests
a8fed99	2026-10-23	crash record / fix	B-spline speed-up, implementation only: layer-1 basis cache for fixed inputs + vectorised exact Hessian (switchable)
201e5ea	2026-10-23	crash record / fix	bspline.py: the mu-calibration stage reads the amended Step 0 (ii) gate (03638f8 item 4); implementation only
9cc112c	2026-10-23	amendment	B-spline: frozen mu = 10^{-6} (dated amendment; growth and certification criteria agree); Step 1 estimate 1.2-3 h typical, 16 h worst case (below the 3-day checkpoint)
d005ee5	2026-10-24	crash record / fix	bspline.py: model keys may carry an explicit P2 mu ('<cond>_P2m<mu>_s<seed>'), implementation only (POST-HOC addendum ff5bd20, check 2)
831f026	2026-10-24	crash record / fix	bspline_posthoc: crash record (thread deadlock in libomp during check 1) and implementation-only fix (one worker process per model)

Changes made after an outcome was known

Generated from the committed files: each entry quotes the committed wording, with the date and commit of the quoted text.

Change	Committed wording	Date	Commit
B-spline replication, calibration step (Step 0)	the fidelity gate, withdrawn	2026-10-22	cba2c07
B-spline replication, calibration step (Step 0)	the polish-ending classification	2026-10-22	4b3f20a

Change	Committed wording	Date	Commit
B-spline replication, calibration step (Step 0)	the analytic null set, null-set-aware certification and certification-based termination	2026-10-23	03638f8
B-spline replication, calibration step (Step 0)	the staleness watchdog	2026-10-23	5bf5acd
B-spline replication, calibration step (Step 0)	the frozen μ	2026-10-23	9cc112c
Dose-response: runtime correlation tolerance	**(a)** The runtime correlation tolerance (0.02) was miscalibrated: it is below the sampling error at n = 2000 (about 0.022 at $\rho = 0$). Pairing and the x_1/x_3 identity passed. Future tolerances are derived from the sampling error of the quantity checked.	2026-10-21	75b778b
Dose-response: added stop reason	**(b)** "Solver error (trust-exact subproblem)" is a dated addition (2026-10-20) to the frozen stop-reason list (crash records: 1d53d85, 11e3663).	2026-10-21	75b778b
B-spline replication: added stop reason (hang)	A polish killed after hanging records "solver error (hang)" and the fact that it was ended by the 2 h + 5 min child kill. This is consistent with the frozen failure-reason list (11e3663).	2026-10-22	4b3f20a
B-spline replication: added stop reason (watchdog)	**Recorded as** "solver error (hang); ended by the staleness watchdog", with the same classification as a hard kill.	2026-10-23	5bf5acd
Synthetic study: X6, pre-registered reading did not fit	**X6's rule maps failure to "non-identifiability persists beyond the runaway". The data contradict that reading.** X6 failed only because the *runaway* cells matched the original teacher *better* than predicted (0.774 and 0.883 against < 0.7). Every cell without a runaway recovered the reference's internals exactly (unit and edge match 1.000, d_AB about 10^{-4}). The verdict stays **failed**, as recorded.	2026-10-03	34cc321
Freeze test: Z2, branch taken and run resumed	By the fixed interpretation, Z2 failing selects the branch "a runaway without composition: the mechanism needs revision; stop and report before anything else". The run was stopped and reported. On 2026-10-17 the author decided, after seeing Arm A's results, to resume Arms D, E and B (option 2).	2026-10-17	9228f70
Freeze test: Z2, pre-registered reading did not fit	**Z2's branch premise did not occur.** That premise is "a runaway without composition". Every Arm A model reached a certified, finite minimum with non-negative curvature (Z3). The failure reflects the growth threshold, which did not separate a distant finite minimum from unbounded growth. The pre-registered rules did not anticipate this.	2026-10-17	9228f70
Applicability analysis: predictive follow-up dropped	The parked predictive study is DROPPED: its pre-stated precondition (off-pattern distance bearing on the fold A polish damage) is not met. Within fold A, the co-solvent off-pattern distance is constant, and every per-row distance is collinear with x_co, through which the damage scales by the model's x_co * h structure. No successor study.	2026-10-25	b19fa2f

E Supplementary figures

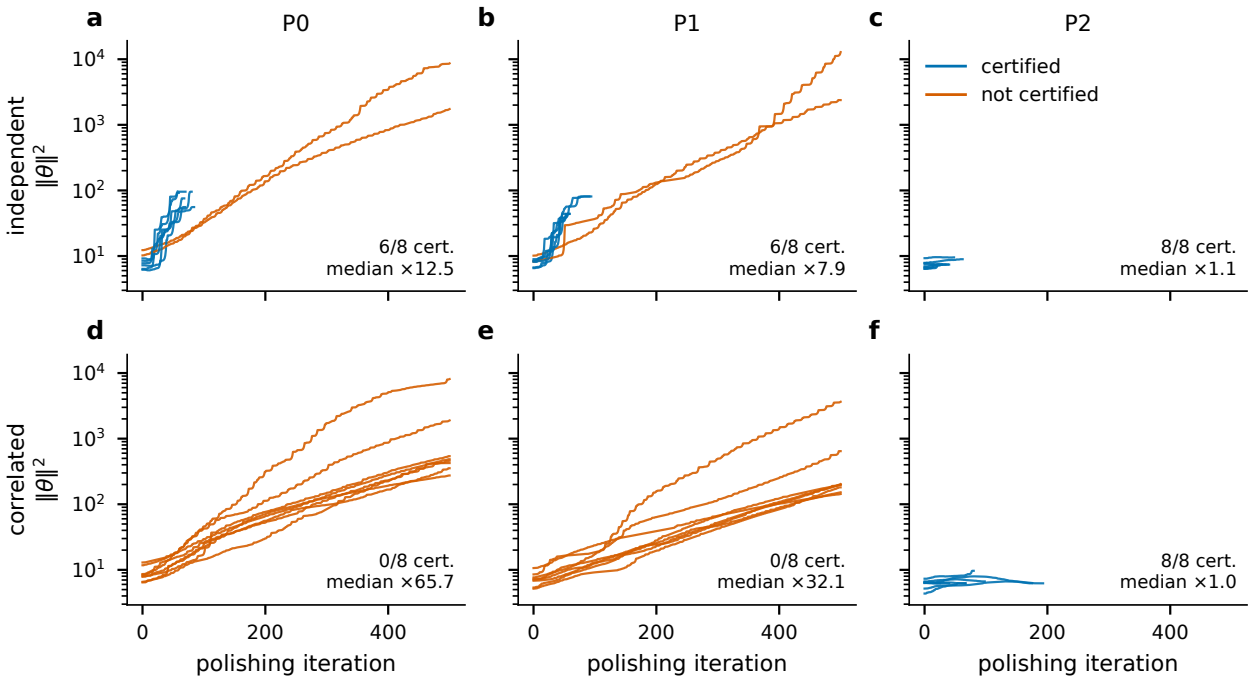


Figure 5: Synthetic 2×3 study: $\|\theta\|^2$ over exact-Hessian trust-region polishing iterations (log scale), one line per seed, coloured certified (blue) or not certified (vermillion). Rows: independent and correlated inputs; columns: P0, P1, P2. Per panel, certified seeds and median growth (polished over early-stopped): independent P0: 6/8 certified, ×12.5; independent P1: 6/8 certified, ×7.9; independent P2: 8/8 certified, ×1.1; correlated P0: 0/8 certified, ×65.7; correlated P1: 0/8 certified, ×32.1; correlated P2: 8/8 certified, ×1.0.

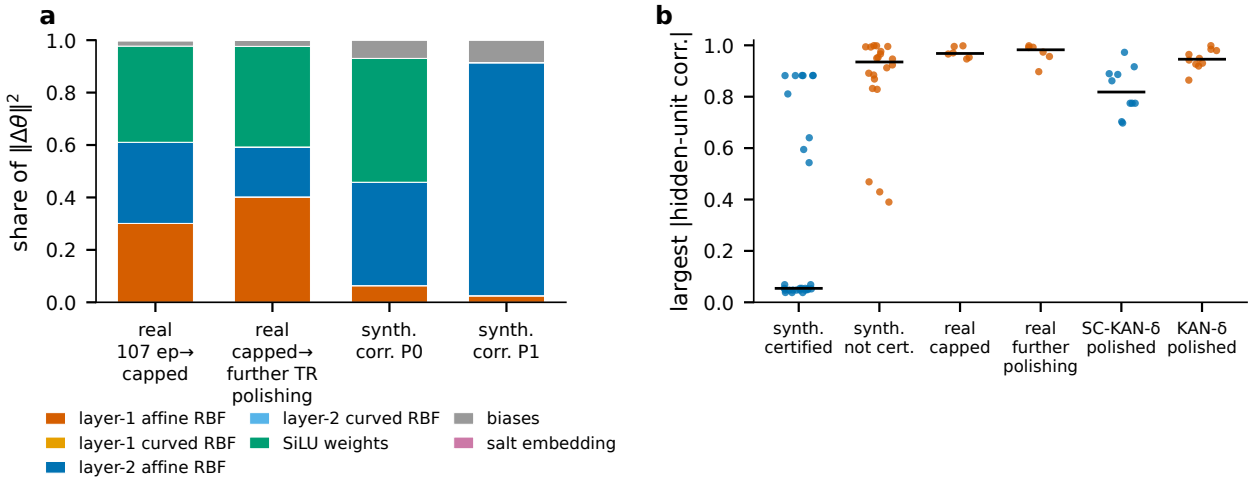


Figure 6: Where the runaway lives. **(a)** Mean share of $\|\Delta\theta\|^2$ by parameter block. Real data, 107 epochs→capped: layer-1 affine RBF 0.30, layer-2 affine RBF 0.31, SiLU weights 0.37; capped→further trust-region polishing: 0.40, 0.19, 0.38; synthetic correlated P0: layer-2 affine RBF 0.39, SiLU weights 0.47; correlated P1: layer-2 affine RBF 0.89. The real-data split was evaluated post hoc from archived weights (Appendix C). **(b)** Largest absolute correlation between hidden-unit activations on the training rows, median: synthetic, certified 0.054 ($n = 28$); synthetic, not certified 0.935 ($n = 20$); real, capped 0.968 ($n = 6$); real, further polishing 0.983 ($n = 6$); SC-KAN-δ polished 0.818 ($n = 10$); KAN-δ polished 0.946 ($n = 10$). Blue: certified; vermillion: not certified or runaway.

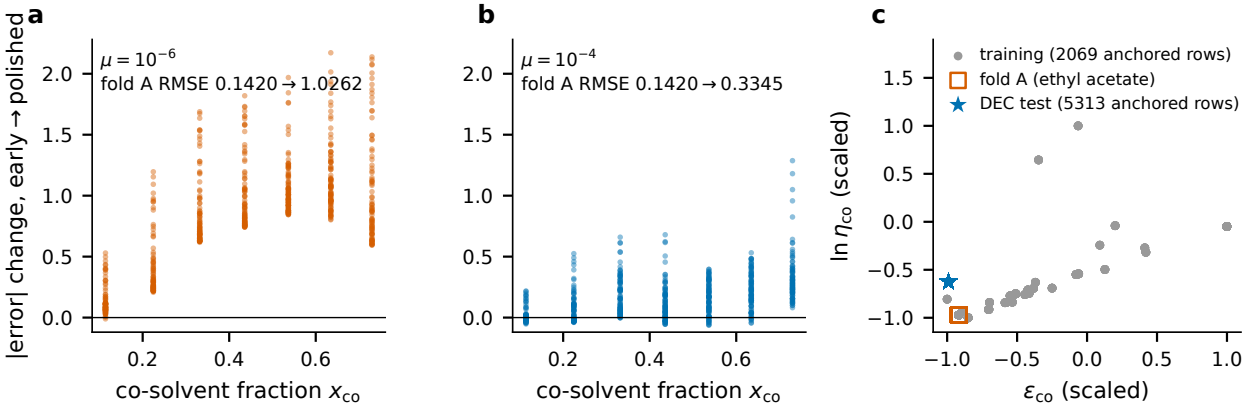


Figure 7: Applicability (descriptive; analysis plan committed before computing). **(a, b)** Fold A (ethyl acetate held out, 635 rows): change in each row's absolute error from the early-stopped 5-seed ensemble to the polished ensemble, against the co-solvent fraction x_{co} (7 distinct values); fold A RMSE 0.1420→1.0262 at $\mu = 10^{-6}$ (a) and 0.1420→0.3345 at $\mu = 10^{-4}$ (b). Per-row values are evaluated post hoc from SHA-verified archived weights, with the logged RMSEs reproduced (Appendix C). **(c)** Co-solvent descriptors ε_{co} and $\ln \eta_{co}$ in the committed min–max scaling: the 2069 anchored training rows (30 distinct points), fold A's co-solvent (off-pattern distance 0.047) and the 5313 anchored DEC test rows (0.334); each held-out set occupies a single point.

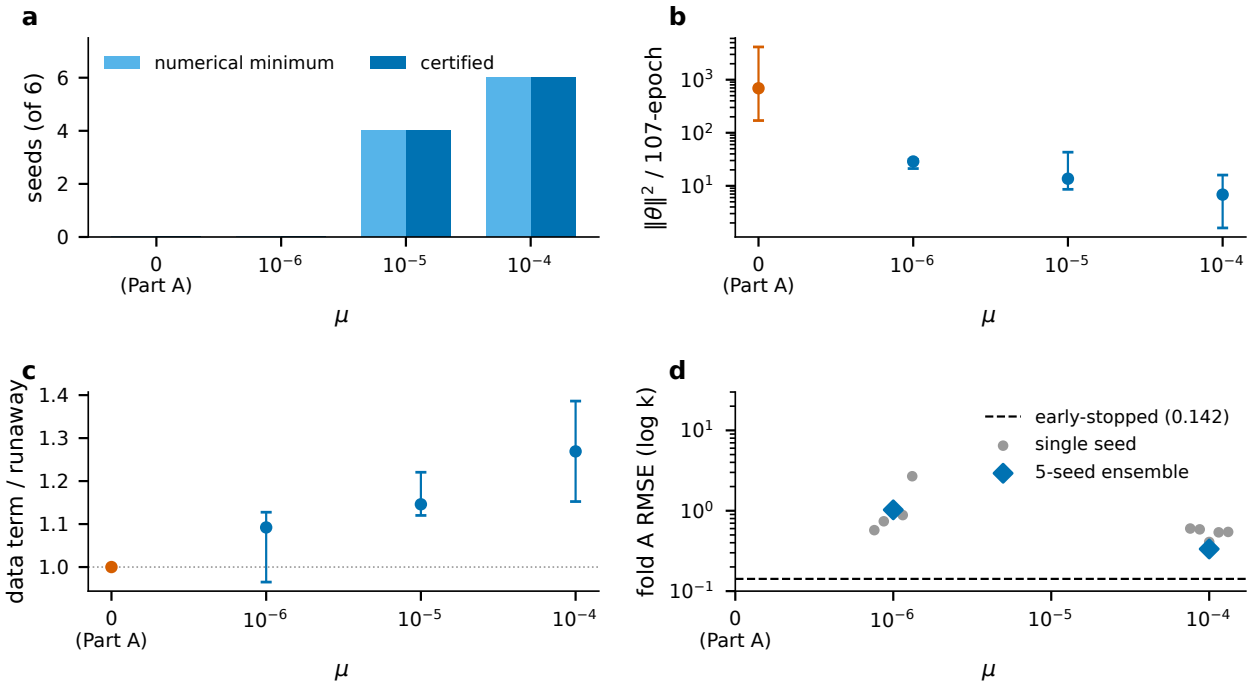


Figure 8: Real-data dose-response in the null-space weight μ (6 KAN-δ seeds; $\mu = 0$: polished without a null-space penalty). **(a)** Seeds at a numerical minimum (all of them certified): $\mu = 0$, 0 of 6; 10^{-6}, 0 of 6; 10^{-5}, 4 of 6; 10^{-4}, 6 of 6. **(b)** $\|\theta\|^2$ relative to the 107-epoch state, median (range): $\mu = 0$, 690 (170–4146); 10^{-6}, 29 (21–31); 10^{-5}, 14 (9–43); 10^{-4}, 7 (2–16). **(c)** Training data term relative to the runaway, median: 10^{-6}, 1.092; 10^{-5}, 1.146; 10^{-4}, 1.269. **(d)** Fold A RMSE: early-stopped 0.142 (dashed line); polished 5-seed ensemble 1.026 at $\mu = 10^{-6}$ and 0.334 at $\mu = 10^{-4}$; grey points are single seeds.

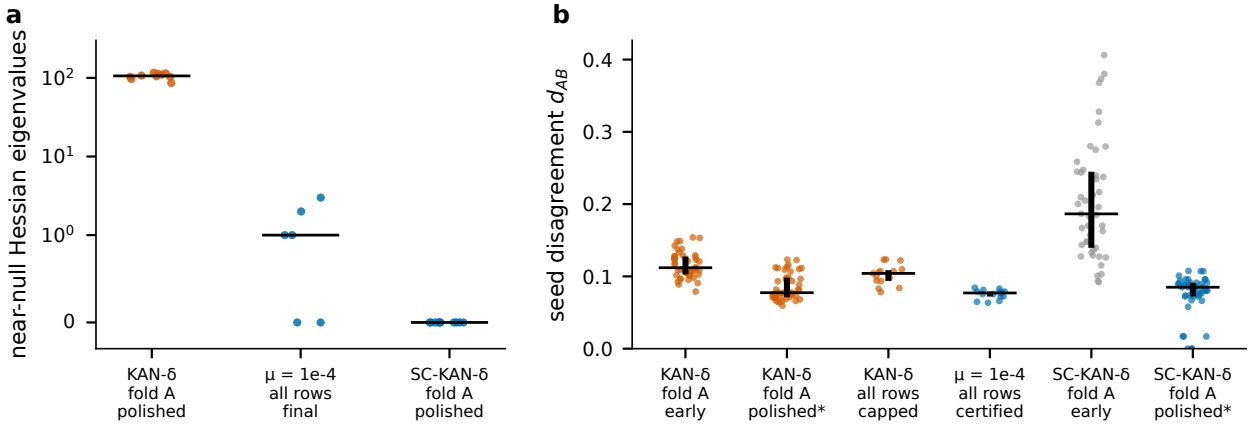


Figure 9: Convergence is not uniqueness. **(a)** Near-null Hessian eigenvalues ($|\lambda| < 10^{-8}\lambda_{\max}$) at the end of polishing: KAN-δ baseline on fold A, median 106 (range 86–116); null-space penalty $\mu = 10^{-4}$ on all rows, median 1 (range 0–3); SC-KAN-δ on fold A, 0 in all 10 models. **The baseline counts are taken at non-stationary points**: none of the 10 baseline models was certified, and all stopped at the 500-iteration cap. **(b)** Seed disagreement d_{AB} (median, interquartile range): KAN-δ fold A early 0.112; KAN-δ fold A polished* 0.077; KAN-δ all rows capped 0.104; $\mu = 10^{-4}$ all rows certified 0.077; SC-KAN-δ fold A early 0.187; SC-KAN-δ fold A polished* 0.085. Groups are evaluated on different row sets (fold A training rows or all rows), so values are not strictly comparable across row sets. Groups marked * (polished fold A) are not in any committed result: they were evaluated post hoc from the archived polished weights (SHA-verified; predictions on the 1434 fold A training rows), without any re-optimisation.

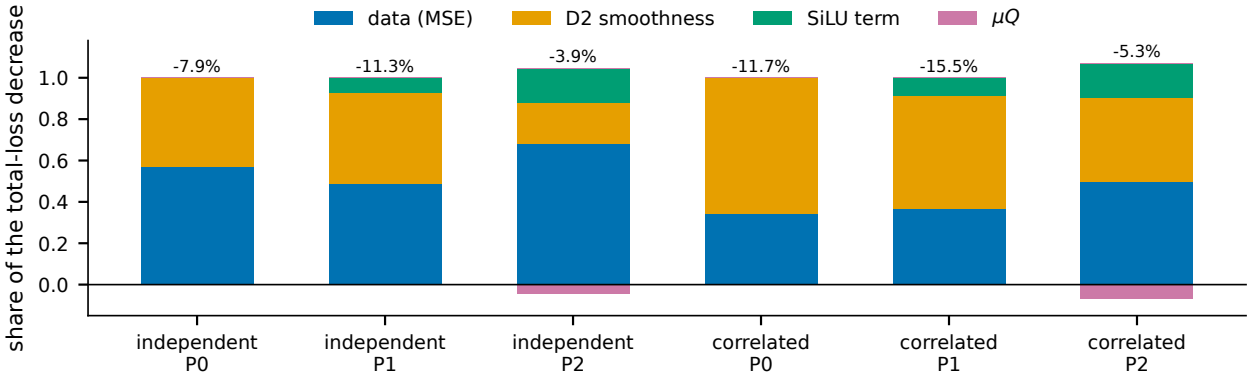


Figure 10: Post hoc loss decomposition (synthetic, early→polished): share of the total-loss decrease from each term, pooled over the 8 seeds of each cell. independent P0: data 0.57, D2 0.43 (median total change -7.9%); independent P1: data 0.49, D2 0.44 (median total change -11.3%); independent P2: data 0.68, D2 0.20 (median total change -3.9%); correlated P0: data 0.35, D2 0.65 (median total change -11.7%); correlated P1: data 0.37, D2 0.55 (median total change -15.5%); correlated P2: data 0.50, D2 0.40 (median total change -5.3%).

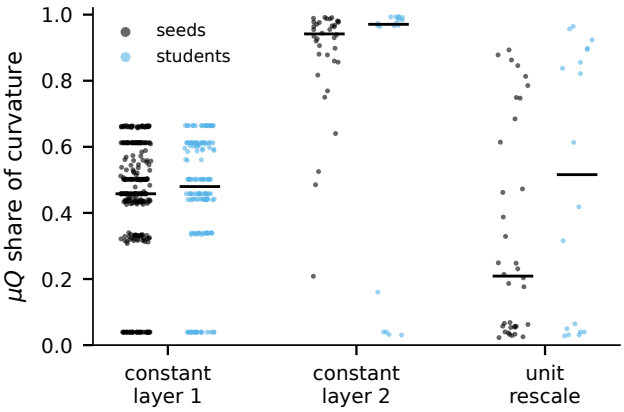


Figure 11: Post hoc gauge check at the $\mu = 10^{-4}$ certified minima: share of the curvature along each least-squares gauge direction contributed by μQ. Medians: constant shift in layer 1, real-data seeds 0.46 ($n = 288$); constant shift in layer 1, students (trained on a frozen teacher) 0.48 ($n = 144$); constant shift in layer 2, real-data seeds 0.94 ($n = 36$); constant shift in layer 2, students 0.97 ($n = 18$); unit rescaling, real-data seeds 0.21 ($n = 36$); unit rescaling, students 0.52 ($n = 18$).

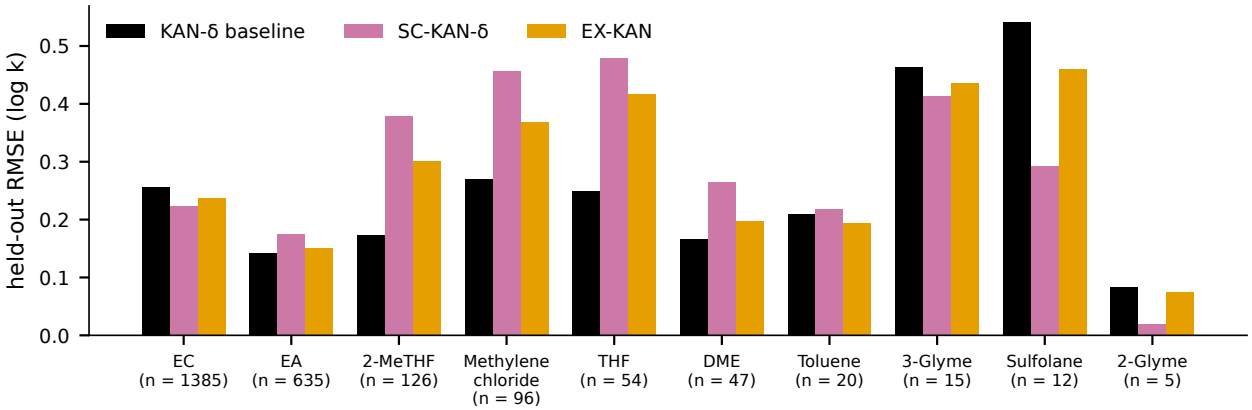


Figure 12: Leave-one-co-solvent-out RMSE per fold (5-seed ensembles, early state) for the KAN-δ baseline, SC-KAN-δ and EX-KAN. Unweighted means: KAN-δ baseline 0.2554; SC-KAN-δ 0.2923; EX-KAN 0.2834. The EA fold uses the fold A rows.

F Supplementary results

These passages were moved from the main text for length; they carry the same evidence and the same labels.

F.1 Freezing the second layer (exploratory arm)

Freezing the second layer instead of the first leaves a non-convex problem. The synthetic models still certified (16 of 16), but the real first layer grew by factors of 10^4 to 10^{10}, mostly through cancellation across its edges, and its endpoints were highly degenerate, with 21–41 near-flat directions; one endpoint that passed the certificate turned out, on a longer restart, to be still drifting outward (exploratory arm; post-hoc follow-up).

F.2 Structured alternatives

Structured alternatives. We tested two architectures designed to remove runaway routes by construction, against the early-stopped baseline, with leave-one-co-solvent-out (LOCO) validation:

Table F.1: Structured alternatives against the early-stopped baseline.

Model	Idea	LOCO RMSE	Converged	Seed agreement
Baseline KAN-δ	As above	0.255	0/10	0.112
Shared-curve KAN	One curve per input, mixed by penalised weights	0.292	10/10	0.187 early; 0.085 polished
Two-anchor excess model	Pure-solvent offset plus $x(1-x)$ mixing term	0.283	not tested	not tested

Seed agreement is the median prediction difference between seeds (lower is better). The shared-curve model converged every time, ending with no near-flat directions where the baseline had a median of 106 (range 86–116, at non-stationary points; Appendix E, Figure 9a), and it was less accurate. At the early-stopped state its seeds agreed less than the baseline's (the pre-registered comparison); after polishing to certified minima their median disagreement fell to 0.085, with some seed pairs reaching the same minimum (post-hoc). The excess model's "pure co-solvent offset" tracked viscosity rather than dielectric constant (rank correlation -0.85, post hoc), reflecting that no training row contains a pure co-solvent. The shared-curve model's learned viscosity dependence also had the wrong sign on more than 10% of rows in 3 of 10 seeds (a pre-registered prediction that failed). Chemically, conductivity should fall with viscosity at fixed composition, but in these data viscosity and dielectric constant rise together (r = 0.966), and higher permittivity favours salt dissociation; with the two nearly collinear, the data cannot separate their partial effects, which is the kind of non-identifiability this paper is about.

F.3 The regularised teacher and gauge fixing

A known solution becomes recoverable. Under the regularised loss, the original teacher is no longer a minimum. Optimising it under the same loss gives a "regularised teacher" computing nearly the same function (correlation 0.9992). Students match its units at 0.87–0.90 and its edges at 0.97–0.98, against 0.68–0.72 and 0.84–0.91 for the original teacher. This comparison was defined after the results and is exploratory.

The penalty acts partly as gauge fixing. At the certified minima, the layer-2 constant shift, an essentially exact symmetry, owes 96% of its curvature to μQ: the regulariser selects one representative. The layer-1 constant shift (46%) and unit rescaling (24%) are only approximate symmetries in this architecture, because Gaussian bumps cannot represent constants exactly and tanh breaks rescaling.

F.4 What a short path cannot tell

What a short path cannot tell. We also predicted that a short regularisation path ($\gamma = 10^{-3}$, 10^{-5}, 10^{-7}), summarised by $D = \|\theta^*(10^{-7})\|^2 \,/\, \|\theta^*(10^{-5})\|^2$, would diverge more strongly as ρ grows. It did not (Spearman 0.71 under P0 and 0.14 under P1, against the required 0.8): D was already high at every dose,

including $\rho = 0$. A post-hoc check, with its classification rule fixed before computing, explains why. Where the unregularised polish certified, mostly at low ρ, 30 of 32 paths were heading towards that certified finite minimum or its valley, none moved outward past it, and 2 found a better minimum at smaller norm. Along one direction with curvature λ, $D = \big((\lambda + 2\gamma_1)/(\lambda + 2\gamma_2)\big)^2$ with $\gamma_1 = 10^{-5}$ and $\gamma_2 = 10^{-7}$, so a finite minimum whose curvature lies within or near the window $2{\cdot}10^{-7}$ to $2{\cdot}10^{-5}$ keeps D large while the path is still approaching it; the observed low-ρ medians (4.54 to 15.05) correspond to $\lambda = 6.7{\cdot}10^{-6}$ to $1.7{\cdot}10^{-5}$. A two-decade D therefore cannot separate a distant finite minimum from a receding one. Receding behaviour (Section 6) is shown only at $\rho = 0.95$ and on the real data, over the tested range; at low ρ the minima are finite but distant; at intermediate ρ these tests do not settle the question. A secondary prediction, that the affine share of the parameter change would rise steadily with ρ, failed.

F.5 The exact symmetry of B-spline edges

Because B-splines sum exactly to one, constants can be moved between a unit's incoming edges and its bias without changing the loss: an exact 15-dimensional symmetry in the [4, 3, 1] model, which holds only approximately for RBF edges. It made the solver hang at converged points and would have made every certified point look degenerate. We excluded these analytically constructed directions, never fitted ones, from certification and from the degeneracy count; exact symmetries are redundancy in the parameterisation, while the runaway directions, which are approximate and data-dependent, remain subject to every check.

F.6 The boundedness criterion

Let L be continuous and bounded below, and $R \geq 0$ continuous and coercive ($R(\theta) \to \infty$ as $\|\theta\| \to \infty$). Then $L + \gamma R$ is coercive for every $\gamma > 0$ and has a global minimiser θ_γ. If L has a finite minimiser θ_0, then $L(\theta_\gamma) + \gamma R(\theta_\gamma) \leq L(\theta_0) + \gamma R(\theta_0)$ and $L(\theta_0) \leq L(\theta_\gamma)$, so $R(\theta_\gamma) \leq R(\theta_0)$ for every γ, and coercivity keeps the θ_γ bounded. Conversely, if the θ_γ stay bounded as $\gamma \to 0$, they have a limit point θ. For any θ, $L(\theta_\gamma) \leq L(\theta) + \gamma R(\theta) - \gamma R(\theta_\gamma) \leq L(\theta) + \gamma R(\theta)$, so by continuity $L(\theta) \leq L(\theta)$: θ is a finite minimiser of L. The path in Section 6 follows certified local minima rather than global ones, so it provides evidence for the criterion's conclusion, not a proof of it.

F.7 Robustness of the regularisation path (post hoc)

Polishing from independent cold starts found no small-norm basin at any γ; at $\gamma = 10^{-7}$ under P1, 8 of 9 cold starts reached a *lower* regularised objective than the warm path, at a larger norm. The warm path therefore understates the divergence. A straight line along the final drift direction rises almost at once: the minimisers recede along a curved valley, not along a ray. The growth sits in the penalty's blind directions, the affine RBF coefficients and SiLU weights of both layers, while the curved coefficients and the salt embedding shrink.

G Methods and definitions

Each term is given in the wording of the file that fixed it, with that file and the commit of the quoted text; code lines are quoted verbatim. *Certified* means that all four criteria hold at the point: a numerically certified local minimum under these tolerances, not a global or unique one. Stop reasons, with solver errors and hangs separated from caps, are tabulated in `paper2/revision/uncertainty_output.txt`.

Term	Committed definition	File	Commit
Edge function (Gaussian RBF + SiLU)	Same edge type: SiLU base plus 8 Gaussian RBFs on fixed centres linspace(−1, 1, 8), width 2/7.	`paper2/synth_prereg.md`	dfd281d
P0 (second differences)	**P0, "D2 only":** $\lambda \cdot \Sigma_e \, \|D2\ c_e\|^2$. This is the final.py smoothness penalty.	`paper2/synth_prereg.md`	dfd281d
P1 (adds the SiLU-weight term)	**P1, "curvature penalty, Bagrow-style":** $\lambda \cdot \Sigma_e$ ($\|D2\ c_e\|^2$ + K_silu · α_e^2).	`paper2/synth_prereg.md`	dfd281d
P2 (null-space penalty)	**P2, "null space closed":** P1 + $\mu \cdot Q$, with $Q = \Sigma_e \, \|P_null\ c_e\|^2$.	`paper2/synth_prereg.md`	dfd281d

Term	Committed definition	File	Commit
Null-space penalty on real data	Loss: the final.py loss + 10^{-4} × Q, with Q from `close.py` (the RBF null-space part of every edge + the squared SiLU base weights).	`paper2/revision/infimum_prereg.md`	349022b
Polishing	**"polished":** trust-exact from the early state, in float64, cap 500 iterations, on the cell's loss. The callback is named `intermediate_result`, with a 2 h safety cap per run.	`paper2/synth_prereg.md`	dfd281d
Certified (definition)	**CERTIFIED** (the certify.py definition, on the cell's loss): max|grad| $< 10^{-7}$; smallest eigenvalue $\geq -10^{-8} \times$ largest; Newton decrement on the non-null eigenspace $< 10^{-16}$; $\|g_null\| < 10^{-9}$.	`paper2/synth_prereg.md`	dfd281d
Certification tolerances (code)	`GTOL, EIGREL, NEWTON, GNULL = 1e-7, 1e-8, 1e-16, 1e-9`	`certify.py`:33	36f295b
Certification criteria (code)	`crit = dict(a=gmax < GTOL, b=bool(ev[0] >= -EIGREL * lmax), c=newton < NEWTON, d=gnull < GNULL)`	`certify.py`:50	36f295b
Near-null directions (code)	`null = np.abs(ev) < EIGREL * lmax`	`certify.py`:45	36f295b
Exact symmetries (B-spline null set)	The null set is the 15-dimensional constant-shift space, constructed analytically from the parameterisation (never fitted from observed flat directions), per the shared-curve Step 0 precedent.	`paper2/revision/bspline_prereg.md`	03638f8
Null-set-aware certification	**Certification of the $\gamma = 0$ training objectives** (data loss + P0 or P1) is null-set-aware: λ_min, the Newton decrement (a complement-restricted solve) and the null-space-gradient criterion are evaluated in the complement.	`paper2/revision/bspline_prereg.md`	03638f8
Degenerate certified point	**DEGENERATE.** A certified point is DEGENERATE if it has more than 3 near-flat directions OR saturation above 10%. (D2: committed synthetic P2 polished models have 0.)	`paper2/revision/infimum_prereg.md`	179814f
Probes at degenerate points	**Three probes at every degenerate point.** All are evaluated on L + $\gamma\|\theta\|^2$ at that γ. 1. **Restart:** trust-exact with the gradient tolerance tightened ×100 ($10^{-7} \to 10^{-9}$), with the same caps. 2. **Scaling:** multiply (i) the layer-1 affine RBF component, and (ii) all layer-1 weights and biases, by c ∈ {1.01, 1.1, 1.5, 2, 5, 10}. 3. **Flat-direction search:** the up to 10 flattest Hessian eigenvectors (smallest $\|\lambda\|$), both signs, loss at 13 log-spaced steps s ∈ {10^-3, 10^-2.75, . . . , 10^0} × $\|\theta\|$ (θ = all parameters).	`paper2/revision/infimum_prereg.md`	179814f
PLATEAU / FLAT MINIMUM	**PLATEAU** if either of these holds:	`paper2/revision/infimum_prereg.md`	179814f
Stalled restart	**Stalled restart.**	`paper2/revision/infimum_prereg.md`	179814f
Path exponent (I2)	**I2.** Runaway models: fit log $\|\theta^*\|^2 = a - \alpha \log \gamma$ over the smallest four certified (incl. certified-degenerate) γ values. α must exceed 0.2, with a bootstrap 95% interval that excludes 0, in $\geq$ 80% of models.	`paper2/revision/infimum_prereg.md`	179814f
Plateau ratio (I3)	**I3.** Control (P2) models plateau: the $\|\theta^*\|^2$ ratio between the two smallest certified γ values is ≤ 1.5, in $\geq$ 80% of models.	`paper2/revision/infimum_prereg.md`	179814f
Short-path statistic D	D = $\|\theta^*(10^{-7})\|^2$ / $\|\theta^*(10^{-5})\|^2$, computed from the points as reached (primary; this is the D that enters R2 and R4). Both points are flagged if either is not certified. A certified-only variant (models with both points certified, incl. certified-degenerate) is reported beside it.	`paper2/revision/dose_prereg.md`	1a18569
Seed disagreement d_AB	d_AB between the 28 seed pairs within a cell: RMS(Δpred) / std(teacher output, training rows).	`paper2/synth_prereg.md`	dfd281d
Unit match	**unit match:** Hungarian on |corr| of unit contributions over the training rows, median over 3 units × 8 seeds;	`paper2/synth_prereg.md`	dfd281d
Cross-edge cancellation ratio	**Logged at both states:** data term, penalty term(s), μQ, $\|\theta\|^2$ per block (layer-1 RBF, layer-2 RBF, SiLU base, biases), the median per-unit cross-edge cancellation ratio (Σ_i RMS(edge_ij) / RMS(Σ_i edge_ij) over the 4 incoming edges, on the training rows), the saturated fraction (|s_j| > 3), max|grad|, and an eigenvalue summary (min, max, number negative $< -10^{-8} \times$ max, null-set size, deciles).	`paper2/synth_prereg.md`	dfd281d

Architecture per experiment, from the code (normalisation layers searched in all 65 tracked Python files: none found):

Experiment	LayerNorm	Biases and interlayer nonlinearity	Code (file:line, committed)
Synthetic study (Gaussian RBF), [4, 3, 1]	none	one bias per unit in each layer (the base Linear); tanh between the layers; no salt embedding; inputs clipped to [-1, 1]	`paper2/synth.py`:55 `class SynthKAN(nn.Module):`; `kan.py`:42 `self.base = nn.Linear(d_in, d_out)`; `paper2/synth.py`:63 `return self.layers[1](torch.tanh(self.layers[0](x)))`
B-spline replication, [4, 3, 1]	none	one bias per unit in each layer; tanh between the layers	`paper2/revision/bspline_basis.py`:76 `class BSynthKAN(nn.Module):`; `paper2/revision/bspline_basis.py`:59 `self.base = nn.Linear(d_in, d_out)`; `paper2/revision/bspline_basis.py`:84 `return self.layers[1](torch.tanh(self.layers[0](x)))`
Real data, KAN-δ [10, 6, 1]	none	one bias per unit in each layer; 2-dimensional salt embedding passed through tanh; tanh between the layers; output multiplied by x_co (delta_head)	`kan.py`:60 `class KAN(nn.Module):`; `kan.py`:42 `self.base = nn.Linear(d_in, d_out)`; `kan.py`:68 `z = torch.cat([x, torch.tanh(self.emb(salt))], 1) # tanh keeps the embedding on the RBF grid`; `kan.py`:70 `z = layer(torch.tanh(z) if i else z) # hidden activations squashed onto [-1, 1] before the next grid`
Shared-curve KAN (SC-KAN-δ)	none	biases in the mixing layer and the head, none in the salt map; tanh after the mixing layer; the shared curves are mean-centred (a fixed centre buffer), which is not a normalisation layer	`arch/sc_model.py`:30 `class SCKAN(nn.Module):`; `arch/sc_model.py`:37 `self.mix = nn.Linear(d, H) # weight = A^T (H x d), bias = b`; `arch/sc_model.py`:39 `self.salt = nn.Linear(2, H, bias=False) # W (H x 2)`; `arch/sc_model.py`:40 `self.head = nn.Linear(H, 1) # w, b0`; `arch/sc_model.py`:53 `return self.head(torch.tanh(self.hidden(x, s, fit)))[:, 0]`; `arch/sc_model.py`:49 `z = g - (g.mean(0) if fit else self.center)`
Two-anchor excess model (EX-KAN)	none	two KAN subnetworks (s: [4, 3, 1]; g: kan.KAN [7, 6, 1] with salt embedding), each with one bias per unit per layer and tanh between layers	`arch/ex_model.py`:33 `class EXKAN(nn.Module):`; `arch/ex_model.py`:24 `self.layers = nn.ModuleList([KANLayer(4, 3), KANLayer(3, 1)])`; `arch/ex_model.py`:27 `return self.layers[1](torch.tanh(self.layers[0](x)))[:, 0]`; `arch/ex_model.py`:37 `self.g = KAN(len(G_COLS), [6, 1], n_salt)`

H Data, splits and chronology

Values below come from `paper2/revision/datasplit.py` (4ed7366; descriptive, no training), which reads the competition files through the committed data path, and from the committed code lines quoted. The competition files themselves are not redistributed.

(a) The PC anchor. The anchor is the measured conductivity of a pure-PC electrolyte with the same salt, from the same source article, at the row's concentration and temperature; it is not neat PC solvent. Per row it is looked up by (source article, salt) among the 13 pure-PC series of the training file, then interpolated: temperature at the series' grid level within 0.3 K, otherwise linearly in $1/T$ between levels (anchored rows: T level 2069); concentration linearly in the article's native unit (c interpolated (linear) 1800, c on grid 269); never extrapolated. Code: `anchor.py`:40 (3081cc0): `def anchor_value(series, c, T):`; `anchor.py`:15 (3081cc0): `T_TOL = 0.3 # K, same grid level`. Anchor information and held-out rows: the pure-PC rows only supply anchors and are in no fold's training or test set. For fold A, the 100 pure-PC rows of fold A's own source article supply its anchors, so a held-out row's anchor comes from the same article as the row. DEC test anchors use same-salt pure-PC series from a fixed list of six training articles, for mol/kg test rows only, averaged where several cover (`anchor.py`:98 (3081cc0): `te_series = {s: [v for (d, salt), v in series.items() if salt == s and d in SAME_SOURCE] for s in TEST_SALTS}`). The input scaler is a min–max map fitted on the anchored training rows and the anchored DEC test rows' inputs (no labels).

(b) Target. $\log_{10}$ of the conductivity k in mS/cm (`log_k` equals $\log_{10} k$ on every training row: True). No conductivity is zero or negative (minimum 1.78e-15 mS/cm); the 45 training rows below 0.01 mS/cm are

dropped (`anchor.py`:70 (3081cc0): `tr = tr[tr.k >= 0.01].reset_index(drop=True)`), and submitted predictions are clipped at $\log_{10} k \geq -2.0$.

(c) Composition. x_{co} is one minus the PC mole fraction, from the file's mole-fraction column; the competition file converts weight- and volume-basis sources to mole fractions (anchored training rows by source basis: w 1779, v 290). Mixtures with more than one co-solvent enter through mole-weighted co-solvent descriptors (dielectric constant, log viscosity, molar mass of the non-PC part) and mole-weighted mixture descriptors (anchored rows by number of solvents: 2 1859, 3 94, 4 116). Molality is native for mol/kg sources and c/ρ_{mix} for mol/L sources, with an ideal-mixing density that neglects the salt (`pipeline.py`:52 (3081cc0): `# molality: native for mol/kg; for mol/L approx c / rho_mix (neglects salt mass)`); anchored training rows by unit: mol/kg 1779, mol/L 290; anchored DEC test rows: mol/kg 5313, all binary.

(d) Row inclusion.

Step (anchor.prepare)	Rows
train rows	6759
test rows	5959
after k filter	6714
known solvents	6520
pure pc rows	1321
pc mixture rows	3214
no pc rows	1985
with same source pc series	2399
anchored	2069
not anchored outside range	330
pure pc series	13
test with pc	5921
test anchored	5313

(e) Folds. Fold A holds out the ethyl-acetate mixtures (EA+PC); the ten co-solvent folds each hold out every anchored row containing that co-solvent, so a row with several co-solvents is in several folds.

Fold	Test rows	Training rows
A	635	1434
2-Glyme	5	2064
2-MeTHF	126	1943
3-Glyme	15	2054
DME	47	2022
EA	635	1434
EC	1385	684
Methylene chloride	96	1973
Sulfolane	12	2057
THF	54	2015
Toluene	20	2049

Fold A: 635 rows; salts LiBOB 635; source articles 10.1149/1.1914757 635; temperature 233.75–332.15 K (11 levels); x_{co} 0.1141–0.7300 (7 values); concentration 0.108–1.016 mol/kg; training rows from the same article: 0. **Fold A is a single salt from a single source article, so salt, source and co-solvent are confounded in it.**

(f) DEC. The test file has no conductivity column (`k`: False, `log_k`: False); all 5959 test rows contain DEC and no training row does. The labels are held by the challenge and were never available to us; this is recorded in the competition's metadata file (`k`: "Absent from test.csv") and in `paper2/revision/appdomain_plan.md` (ec769ad): "The DEC test labels are hidden". No DEC number in this paper is an accuracy.

(g) Selection chronology. The 107 epochs used by every KAN-δ model were chosen on held-out folds: in `kan.py` (3081cc0) each of 5 seeds was early-stopped on fold A and on a second fold (EC+PC) using that fold's own held-out rows (`kan.py`:209: `for s in SEEDS: # early stop on this fold (as specified; optimistic)`), and the median of the 10 best epochs was fixed for refits. The committed log (`kan_output.txt`, 3081cc0): `A: PC+EA done (KAN-delta best epochs [103, 111, 138, 73, 129], VTF-KAN best epochs [21, 20, 43, 26, 22]); B: EC+PC done (KAN-delta best epochs [4, 116, 85, 31, 111], VTF-KAN best epochs [400, 145, 399, 400, 400]); median best epochs: KAN-delta 107, VTF-KAN 94`. **Fold A was therefore reused during development: it informed the epoch count, and the committed studies below evaluated on it before the paper's final comparisons.** The list is every committed output or pre-registration that reports on fold A, by the commit that added it.

Date	Commit	File	Commit subject
2026-09-23	3081cc0	`anchor_output.txt`	CALiSol-23 DEC holdout: EDA, GBM baseline, anchor + KAN-δ model, final submission
2026-09-23	3081cc0	`compare_output.txt`	CALiSol-23 DEC holdout: EDA, GBM baseline, anchor + KAN-δ model, final submission
2026-09-23	3081cc0	`extend_output.txt`	CALiSol-23 DEC holdout: EDA, GBM baseline, anchor + KAN-δ model, final submission
2026-09-23	3081cc0	`kan_output.txt`	CALiSol-23 DEC holdout: EDA, GBM baseline, anchor + KAN-δ model, final submission
2026-09-23	3081cc0	`kanfull_output.txt`	CALiSol-23 DEC holdout: EDA, GBM baseline, anchor + KAN-δ model, final submission
2026-09-23	3081cc0	`pipeline_output.txt`	CALiSol-23 DEC holdout: EDA, GBM baseline, anchor + KAN-δ model, final submission
2026-09-23	c11dd92	`gauge_prereg.md`	Pre-register KAN identifiability study (before any experiment)
2026-09-23	e68e2eb	`gauge_output.txt`	Identifiability study: gauge.py, control result (failed), verdicts
2026-09-29	175478a	`electrolyte_case_study_summary.md`	Close-out: electrolyte case study summary (all pre-registered predictions)
2026-09-29	6c93bf4	`rule_prereg.md`	Pre-register basis-size rule intervention (before rule.py exists)
2026-09-29	df44ead	`rank_prereg.md`	Pre-register rank-constrained layer-1 intervention (before rank.py exists)
2026-09-30	aeb6f87	`rank_output.txt`	Rank intervention: results (I1, I2 failed both arms; I3 passed; I5 held)
2026-10-02	e5105d7	`close_prereg.md`	Pre-register targeted null-space penalty intervention (before close.py exists)
2026-10-03	0053d01	`arch/sc_prereg.md`	arch: pre-register SC-KAN-δ (shared-curve KAN-δ) before any script exists; decisions C (centring), mu ...
2026-10-03	36f295b	`certify_output.txt`	Certify: W1, W5 held; W2, W3, W4, W6 failed; certified minima are isolated, non-convexity remains
2026-10-03	5f4520a	`certify_prereg.md`	Pre-register second-order certification and regularised fold A accuracy (before certify.py exists)
2026-10-03	65ed55f	`close_output.txt`	Close (null-space penalty): V1, V6 failed; V2-V5 inconclusive; numerical stationarity and fold A overfitting n...
2026-10-03	d898d73	`arch/sc_output.txt`	arch SC-KAN-δ Step 0: unit tests ok, LOCO table, beta frozen at 10^{-4} (10^{-5} failed: median x7.66), certific...
2026-10-04	3938f8f	`arch/ex_prereg.md`	arch: pre-register EX-KAN (two-anchor excess model) before any script exists; decision B (x_co-independent g, ...
2026-10-04	4c268c1	`arch/ex_output.txt`	arch EX-KAN Step 0 data checks (no training): 3 co-solvents flagged (2-Glyme, 3-Glyme, Sulfolane), 7 unflagged...
2026-10-04	6e1df32	`paper2/checks/checks_output.txt`	paper2/checks (POST-HOC, descriptive): hidden-unit correlation marks the runaway (independent cells 0.70 vs 0....
2026-10-16	a6d752b	`paper2/revision/freeze_prereg.md`	Revision step 2: pre-register the composition test (frozen layer 1, one-layer control) and the weight-decay ar...
2026-10-16	fcda1c1	`paper2/revision/checks_output.txt`	Revision step 1 (descriptive checks and code hygiene): DEC-gain claim source, AdamW decay sizing, penalty norm...

Date	Commit	File	Commit subject
2026-10-17	11bf8bc	`paper2/revision/freeze_output.txt`	Revision step 2: decision record - freeze.py run stopped by the pre-registered rule (Z2 failed in Arm A); the ...
2026-10-25	656b06c	`paper2/revision/appdomain_output.txt`	Applicability domain (DESCRIPTIVE): fold ensembles reproduced (0.1420 / 1.0262 / 0.3345); off-pattern tracks L...